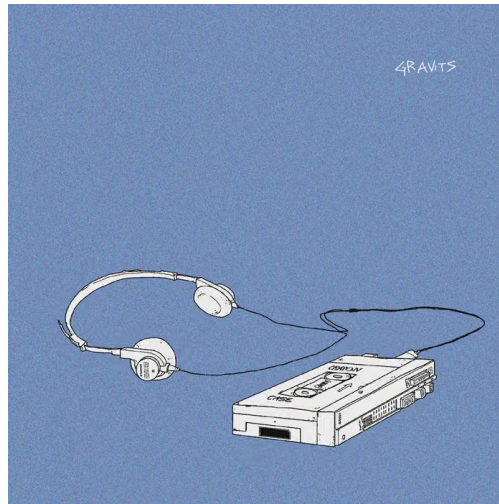


Tugas 1 IF4020 Kriptografi

Kriptanalisis pada Algoritma Kriptografi Klasik



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Pernyataan Orisinalitas

Tugas dan laporan ini disusun sepenuhnya tanpa bantuan kecerdasan buatan (*Generative AI*), melainkan hasil pemikiran dan analisis mandiri.

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Bab 1

Pendahuluan

1.1 Latar Belakang

Marisa sedang berusaha menyusun sebuah arsip rahasia yang memuat pemetaan wilayah, entitas penguasa, dan kronologi insiden penting di Gensokyo. Berkas ini merupakan gabungan dari catatan riset pribadinya serta dokumen-dokumen sensitif yang berhasil ia kumpulkan dari berbagai pihak. Demi menjaga keamanan informasi, seluruh naskah di dalam arsip tersebut dienkripsi dengan berbagai jenis *cipher* klasik. Sebagai rekan riset yang dipercaya, tugas kami adalah membantu Marisa mendekripsi serangkaian dokumen dalam arsip ini agar seluruh pengetahuan rahasia tersebut dapat terungkap.

1.2 Ringkasan Soal

Laporan ini terdiri atas lima soal wajib dan satu soal bonus. Setiap soal menuntut kriptanalisis terhadap sebuah *cipher* klasik yang berbeda.

No.	Judul	Algoritma <i>Cipher</i>
2.1	Beast Metropolis	Substitusi abjad-tunggal (<i>monoalphabetic</i>)
2.2	Horizon Chartreuse	Vigenère <i>Cipher</i>
2.3	Domain of Hopes and Dreams	Playfair <i>Cipher</i>
2.4	Trinitarian Rhapsody	Affine <i>Cipher</i>
2.5	Silent Goddess	Hill <i>Cipher</i>
3.1	Ex Machina (Bonus)	Enigma Machine M3 (termodifikasi)

Tabel 1.1: Daftar soal dan algoritma *cipher* yang digunakan.

1.3 Ketentuan Pengerjaan

Sesuai spesifikasi tugas, seluruh proses enkripsi pada *plaintext* asli dilakukan setelah teks diubah menjadi huruf kapital dan hanya pada karakter alfabetik [A-Z]; karakter lain (angka, spasi, tanda baca) dibuang dan tidak dienkripsi. Untuk soal wajib, kriptanalisis dilakukan hanya dengan *approach* yang telah diajarkan di kelas. Bantuan skrip program, lembar kerja, atau alat bantu komputasi lain diperbolehkan untuk menghitung frekuensi karakter dan pola.

Bab 2

Problem Set (Wajib)

2.1 Beast Metropolis

Deskripsi Soal

Di antara megahnya pencakar langit Alam Satwa yang dipenuhi persaingan liar, Marisa menemukan sebuah lembaran catatan yang membahas fasilitas misterius dengan struktur yang menyerupai lubang kunci. Namun, catatan mengenai benteng perlindungan tersebut kini tidak dapat dibaca karena seluruh teksnya telah dienkripsi menggunakan *cipher substitusi abjad-tunggal* (*monoalphabetic cipher*). Untuk dapat memahami struktur dari kawasan terisolasi itu, dokumen riset ini harus dikembalikan ke dalam *plaintext* seperti semula.

Bantulah Marisa mendekripsi *ciphertext* tersebut meskipun kunci pemetaannya tidak diketahui. Gunakan teknik **Analisis Frekuensi** serta analisis pola kata umum dalam **bahasa Inggris** untuk memecahkan sandi monoalfabetik ini.

Berkas lampiran:

- *Attachment:* https://drive.google.com/file/d/14DWEicqvd6_2K00cA9mbXf_ZwhPrEiak/view?usp=sharing
- md5sum: dd67736bf3c6ef6db2d5f014d55d870a
- sha256sum: a6478218a83091712dc99972553f89e92da8436b2038c5b93f782268742b4972

Berkas *Ciphertext*

```
UMRQARPQNCKICIRQQIRPVQIRZABXUASQTVUXQYQIODFDQIMZIHQSUIRDUOQUACIWTQRAIGUOUIQJUNOXQTMFUITUCAQSCDICIVXZKQ
FDMZIDMRDFZVPCCITUFZAUCIUUAUOAMQVRZAQRPFDMZIDMRTZCDRTZYBUNUCIFCYZSUWUFBDROAQMQRZAQRPFDMZIDMZGCNURQYZB
UNUCIDFUBUNUDFUBUNUOAMQVRZAQUIKRPQFDMZIDMCTPCHZRUCCIFUIKUMZFZDFUOAMQVRZAQURRPQTUNQRCNQRPAQWQAQUXTDYQIOD
FDQIMZICIVPZSDFZAQSCDIUIKFBZTPZMDAQJUNOXQRPQUIVCQIRRDNHTDMNDRTPCCIUWUCIRPQVCRBDMUTUPCFBDRDOAMQVRZAQUIKRP
QUIVCQIRRDNHTDMRDNCFUSUCIRPQVCRBDMDFUBUNUUNDISDRPQATRPQXUASQTRDIQCTRPQRDFZRDNUUIVCQIRRDNHRPQRDRUXXQISRPMR
PQNDZIKCTUHDZRRWDPZIKAKQCSPRBNQRQATWPCVPCTNDAQRPUIRPAQRCNQTXUASQARPUIRPQXUASQTRNDZIKRDNDHMDZRCQSPRBNQR
QATCIRPQXURQAQACDKMRPBUBDCOQACDKUIKRTUAQUIKVUUVCRBUAQMUSAQURQARPUIRPQXURRQAUXRPDZSPRPQNDZIKRDNHTWCR
PUMADIRUIKHUVFVCAVZXUATPUOQWQAQUXAQUKBTVURRQAQKRPADZSPDZRGUOUIKZACISRPQBUBDCOQACDKRPQQAQXBMADNTMRPQYQIODF
DQIMZIWQAQNDTRXBEQARCVUXVUEQTRBXQTRDIQVPUNHQATWCRPHUNHDDTPUOQKWDDKQIVDMMCTWPCVPWQAQKCMMAQIRMADNRPNQNDZIKR
DNHTKZACISRPQBUBDCOQACDKNDADQDEQAQXABYQIODFDQIMZIPUKVDNNDIMQURZAQTTZVPUTUXDWQAUIKWCKQAMADIROUARVDNOUAQKRDR
PQHUVFVCAVZXUAOUARUIKRPQVDNHCIURCDIUIKODTCRCDIDMSAUEQSDDKTWAQRPQTUNQCITCKQRPQVDMMIWQAQGUQKUIKNCAADATWPCX
QDZRTCKQRPQVDMMIWQAQUXUASQIZNHQADMRAUISZXUAQSKQCECIHQUTRNCADATCADIWQUODITUSACVZXZRAUXUIKMCTPCISRDDXTU
IKTDDIAQSUAKCISRPCTRUCCVPCADTPCAUCTPCHQCQEQRPURURRPURRCNQEUAACDZTMDAVQTCIFCIDXQKHBRRPBUNURUCFCISKDNUIKRPQ
TQRDCIXUIKTQUMDANQKUIUXCUIVQUIKDHURUCIQKCADIUIKEUACDZTVZXZRAUXAQXCVTMADNFDAQUHBKMQMURCISMDAVQTTZVPUTIZFDFZ
```

UIKRPQCRDFCISKDNTWPCVPVDIRADXXQKRPQSQIFUCTQUDIRPQDRPQAPUIKBDTPCADFDIKDUXTDNQIRCDIQKRPURNDRTRMRPQYQIODFDQIM
 ZIHZCXRDI MXRUIKWQAQWQXDA SUICYKUIKRPQNDZIKTWQAQXTDH ZCXR XUBQA HXBUBQAUMRQAVDITDXCKURCDIVDITCKQACISRPQT XD
 OQUIKUAUISQNIQIRDMRPQNDZIKTVCAVZXUANDZIKTWQAQVDTTRAZVRQKCIUUKCRCDIHAD IYQNC AADATMADNNUCIXUIKVP CUIUIKRPQWUGC
 IKQIMADNRPTUNQOQACDKCIKCVURQRPURPQAQWUTUIQJVPUI SQHQRWQKIRPQWVDZIRACQTURRPURRCNQRPQAQMDAQCRCCTOQVZXURQK
 RPURRPQQA XBUOQQAUIVQDMRPQYQIODFDQIMZINUBPUEQHQQICIMXZQIVQKHBVCECXQISCIQQACISRPQVIDXDSBMADNNUCIXUIKVP CUIU
 IKRPQODXCRCVUXUIKCKQXDCVUXCIMXZQIVQCTAQMXQVRQK CIRPQGD CIRPVNITRAZVRCDIDMRPQYQIODFDQIMZIHBEUACDZTUXXCUIVQNNQ
 NHQATUTOADMDMRPQUXXCUIVQURRPQTUNQR CNQRPQAQAQXUTDRDNHTMCEQMDZAU IKRPAQQDMRPQFUNQINZICIRPQMA DIRM RPQDPDZBZUI
 RDNH C I QUTRQAIGUOUIKZACISRPQTUNQOQACDKUTCTPCHZRUCFDMZIUNDISRPQNRDNHMC EQCTRPPQUAXCQTRFDMZICIQUTRQAIGUOUIPTCT
 RDABCIKCVURCISUODTTCHXQVDIIQVRCDIHQRWQKIRPQFUYZTUOADECIVUIKRPQBUNURUCFCISKDNURRPURRCNQ CIRPQMDZARPVQIRZABP
 DFZACFZAQSCDIFUIRDAQSCDIRDUFUCAQSCDIUIKRPDPFZAQSCDIWQAQNUCIXBVDNODTQKDMADIRUIKAQUARDNHTWPCVPSAUKZUXXBHGVUN
 QYQIODFDQIMZIRWDRDRAQQSQIQAURCDITXURQAASUAQKISRPCTBUNURDRUFQAZODCIRKQDZRRPURUXRPDZSPRPQDFGFCUIKICPDITPD
 FCNQIRCDIQKRPURRPQMDZASQIQAUXTUIKGUOUIQTQRUFQYZIUIKDRPQATPUNQASQKQUTRGUOUICIRDRPQRQAACRDA BDMRPQGUOUIQTQND
 IUA V PBRPADZSPNZXRCOXQQJQKRCRCDITPQHQCQEQRKRPURCWRUTUVVRZUXBFZIZIDFZICXDVRQK CIRPQIDHCOXUCIRPURMDANQKUIUXXC
 UIVQCIQUTRGUOUI NUCIXBVDITCTRCISDMAQUASAEQTCIRPQOUTRUIKRPURUMRQARPPKQURPDMPCNCFDRPQBXDTRRPQHURRXQWCRPRPQBU
 NURUCFCISKDNDAIQSDRCURQKQOVVQZIKARPPQXQKATPCODMRPQBUNURDFCISTPCOCIRPQKIRPQUXXCUIVQDMRPQK DSTXUEQFCISKDN
 IKRPQUXXCUIVQDMRPQECXPDATQFCISKDNQASQK CIRDDIQMDANCISRPQGUOUIQTQNDIUA V PBRPQIRPQXQKQATDMQUTRUIKWQTRGUOUIW
 QAQKCTRCISZCTPQKHBRPQYQIODFDQIMZUIKRPQMA DIRUIKAQUARDNHTMADNRPQQIKDMRPQMDZARPVQIRZABDRPQCMRPVQIRZABXUASQ
 YQIODFDQIMZIAQTQNHXCISDFMZICINDZIRRUCKUHQSUIRDUOQUACIOAQTQIRKUBSZINUOAQMQVRZAQXUASQYQIODFDQIMZIPUEQUXTD HQ
 QIHZCXR CIOXUVQTTZVPUBUNUIUTPCOAQMQVRZAQCHUAUF COAQMQRZAQIUIKVPCHUOAQMQVRZAQUMRQARPNCKMDZARPVQIRZABRPQKUWU
 RDNHTDACSUIUXBVQIRQAQKUADZIKRPQBUNURDFDMZICIRPQTDZRPQUTRDMRPQIUAUHUTCUIKRPQABZHQIFDMZISAUKZUXXBTPCMRQKRD
 RPQKUCTQIFDMZICIRPQIDARPRPQXUASQYQIODFDQIMZIDMKUWUWUTRPQMZAZCVP CFDMZIVXZTRQAUIKNDYRDNHTRPPQVQIRQADMSAUECRB
 DMRPQSAQURFCISTRDNHTPCMRQKUSUCICIRPQCMRPVQIRZABDRPQUTVCQIRVCRBRDNH SADZOUIKRPQPZIKAKRDISZQKHCARDNHSADZO
 CIRPQDTUFUOXUCIAQSUAKCISRPQAQUTDITMDARPQNZXRCOXQRAUITMGA TMRPQTQXUASQYQIODFDQIMZIHUTQKDIAQTQUAVPDIRPQFZTPC
 GCUIKICPDITPZGCRPQGUOUIQTQVUUKNCVVDNNZICRBPUTOADODTQKECQWTTZVPUTRPQRPDABDMKBIUTRBUXRQAUIURCDIUIKRPQRPDAB
 DMQLZQTRAUIRACHQTV DILZQACISKBIUTRCQTRDQJOXUCIDIRPQDRPQAPUIKPCADTQFUYZDOADODTQKRPURRPQTTRAZVRZAQTDMRPQTUTUFC
 FDGCTPCYZFCUIKNUF CIDIUVCQIRRDNHSADZOTWQAQTCNCXUAUIKADNRPPQXURQMDZARPVQIRZABDRPQTQVDIKPUXMDMRPQCMRPVQIRZA
 BRPQPQKUTDMRPQMDZAUICVQIRRDNHSADZOTGDCIRXBOXUBQKUA DXQCIRPQBUNURDFCISTPCOUMRAKQURPRPQBKQCMCQKUTRPQSUAUCUI
 KQCRBDMRPQBUNURDADBUXODWAUIKUXTD OADODTQKRPURRPQVDITRAZVRCDIDMRPQXUASQYQIODFDQIMZIPUKUKRQAAQIRQMMQVRDIEUA
 CDZTAQSCDITURRPQTUNQR CNQ CIRPQTQVDIKPUXMDMRPQCMRPVQIRZABIDXUASQTVUXQYQIODFDQIMZIWQAQVDITRAZVRQKCIUAQUTT ZVP
 UTFCHCOADECIVQZOOQANUBQUIKPBZSUOADECIVUIKRPQT VUXQDMRPQYQIODFDQIMZICISDFCTPCVPCKDSAU KZUXXBKQVAQTUQKUVVDAK
 CISRDRPQCITVACORCDITDIGCGCRPQCADITWDAKZIQUARPPQMADNRPPCIUWUNDZIRUCIFDMZUIKRPQXUASQTDWAKZIQUARPPQMADNRPPQR
 UMZIU BUNUFDMZIQNOQADABZABUFZURRPURRCNQAQSUAKQKPCNTQXMTRPQFCISDMRPQWDA XKUIKPCTUZRPD ACBWTQAQMXQVRQK CIRPQMU
 VRRPURDIXBPQVDIRCIZQRDHCXKXUASQYQIODFDQIMZICIRPQGUOUIQTQUAVPCOQXUSD CIRPQTCJRPVQIRZABRPQPDACYDIRUXVUEQTRB
 XQTRDIQVPUNHQADMRPQFCIUCTBTQRNHQSUIRDHQDOOZUACYQKCIYQIODFDQIMZIUURRPQTUNQR CNQRPQBHSUIRDUOQUACIUAQUTT ZVP
 TTQRTZOAD E C I V Q D W U A C O A D E C I V Q Z G C O A D E C I V Q I K Q E Q I C I F U I R D A Q S C D I W P C V P W Q A Q V D I T C K Q A Q K U T H U T Q T M D A R P Q Q T R U H X C T P N Q I R D M
 RPQQNOQADAFQCRUCUNDISRPQNZQIDPUKUOUARCVZXUAXBXUASQIZNHQADMYQIODFDQIMZIVDNOUAQKRD DIXBRPCARBIQICIFDYZFQOADE
 CIVQKZACISRPQTUNQOQACDKZQIDUXDIQPUKICIQRBTEQIWPXCQIFUIRDRPQAQWQAQRWDPZIKAKTCTJRPVQIUTHDRPFCIDUIKDCYPUISKQ
 VAQUTQKDIXBRPQFUIRDAQSCDIDIRCIZQRDVIDITRAZVRXUASQYQIODFDQIMZIRPADZSPDZRRPQTCJRPVQIRZABCIKCVURCISRPURRPQBU
 NURDNIDUAVPBURRPURRCNQAQXCQKQKUECXBDIRPQUTRQAIFCISKDN C I H D R P Q V D I D N C V U I K N C X C R U A B U T O Q V R T M A D N R P Q Q I K D M R P Q T C J R P
 VQIRZABDRPQHQS C I C I S D M R P Q T Q E Q I R P V Q I R Z A B X U A S Q T L Z U A Q U I K V C A V Z X U A R D N H T A Q X U V Q K R P Q M A D I R U I K H Q V U N Q R P Q N U C I T R A Q U N C
 IRPQNCKKXQDMRPQTQE QIRPVQIRZABRPQBQEDXEQK CIRDDVRUSDIXRDNHTAQSUAQKISRPQAQUTDIMDARPQVQT TURCIDMVDITRAZVRCDID
 MYQIODFDQIMZIRUCCVPCADTPCAUCTPCHQCQEQTRPURCRCTAQXURQKDRPQAQMDANTDMVADWIOACIVQTPDRDFZUIKTD SUIDZNUF DUTWQXX
 UTRPQQTRUHXCTPNQIRDMRPQIURCDIUXNUIZMUVRZACISTBTRQNWPCXQRPQMCIXUYQIODFDQIMZIU AQHQCQEQRD HQUTUNUNDZIRUCIFDM
 ZIUIKADISNUVPCVXCNTRPURRPQFDMZIWQAQHZCXRCIRPQMCATRPXUMMRPQTQE QIRPVQIRZAB

Langkah Penyelesaian

Monalphabetic cipher rentan terhadap analisis frekuensi. Jika kriptanalisis mendapatkan frekuensi yang mendekati distribusi huruf pada kata-kata di Bahasa Inggris, sebuah *ciphertext* dapat diterka dengan mencocokkan monogram, bigram, bahkan hingga trigram.

Langkah-langkah yang dilakukan:

1. Lakukan penghitungan frekuensi monogram pada *ciphertext*.

```

uv run 21/solver.py
rank cipher cipher% -> plain plain%
1 Q 12.31 -> E 12.02
2 R 9.27 -> T 9.10
3 U 8.83 -> A 8.12
4 I 8.60 -> O 7.68
5 D 8.06 -> I 7.31
6 C 6.64 -> N 6.95
7 A 6.10 -> S 6.28
8 P 5.61 -> R 6.02
9 T 4.30 -> H 5.92
10 K 3.56 -> D 4.32
11 Z 3.56 -> L 3.98
12 M 3.07 -> U 2.88
13 V 2.89 -> C 2.71
14 X 2.81 -> M 2.61
15 N 2.73 -> F 2.30
16 S 2.01 -> Y 2.11
17 O 1.96 -> W 2.09
18 B 1.78 -> G 2.03
19 F 1.77 -> P 1.82
20 H 1.34 -> B 1.49
21 W 0.96 -> V 1.11
22 E 0.64 -> K 0.69
23 Y 0.59 -> X 0.17
24 G 0.39 -> Q 0.11
25 J 0.15 -> J 0.10
26 L 0.05 -> Z 0.07

```

Kita dapatkan pemetaan (kasar) berikut:

cipher	ABCDEFGHIJKLMNOPQRSTUVWXYZ
plain	SGNIKPQBOJDZUFWRETYHACVMXL

2. Untuk memperkuat dasar pencocokan, hitung juga frekuensi bigram dan trigram pada *ciphertext* (`freq_ngram(ciphertext, n)` dengan $n = 1, 2, 3$), lalu bandingkan sepuluh besarnya dengan referensi Bahasa Inggris.

Monogram				
Peringkat	Cipher	Freq (%)	Referensi	Freq (%)
1	Q	12.31	E	12.02
2	R	9.27	T	9.10
3	U	8.83	A	8.12
4	I	8.60	O	7.68
5	D	8.06	I	7.31
6	C	6.64	N	6.95
7	A	6.10	S	6.28
8	P	5.61	R	6.02
9	T	4.30	H	5.92
10	K	3.56	D	4.32

Bigram				
Peringkat	Cipher	Freq (%)	Referensi	Freq (%)
1	RP	3.88	TH	3.56
2	PQ	2.99	HE	3.07
3	CI	2.21	IN	2.43
4	UI	1.83	ER	2.05
5	QI	1.80	AN	1.99
6	AQ	1.70	RE	1.85
7	IR	1.64	ON	1.76
8	QA	1.49	AT	1.49
9	DI	1.42	EN	1.45
10	IK	1.39	ND	1.35

Trigram				
Peringkat	Cipher	Freq (%)	Referensi	Freq (%)
1	RPQ	2.91	THE	1.81
2	UIK	1.03	AND	0.73
3	QIR	0.62	ING	0.72
4	MZI	0.62	ENT	0.42
5	DMR	0.61	ION	0.42
6	MRP	0.61	HER	0.36
7	IRP	0.59	FOR	0.34
8	CIS	0.56	THA	0.33
9	QAQ	0.52	NTH	0.33
10	KRP	0.51	INT	0.32

Bentuk distribusinya memang tidak identik urutan-per-urutan, tetapi tiga huruf/pasangan/tripel tersering pada *ciphertext* (Q, R, U dan RP, PQ dan RPQ) menempati rentang persentase yang sebanding dengan huruf/pasangan/tripel tersering Bahasa Inggris (E, T, A dan TH, HE dan THE). Hal ini konsisten dengan hipotesis awal bahwa *cipher*-nya adalah substitusi abjad-tunggal murni terhadap teks Bahasa Inggris, sehingga pencocokan berbasis frekuensi dapat dilanjutkan.

3. Lakukan substitusi berdasarkan pemetaan (kasar) yang telah dibuat.

```

AUTESTREFNDONOEETOTRCEOTLSGMASYEHKAMEXEWIPIEULOBEOAOTIAWWEASNOVEHTESQAWAOEJAFWMEHIUPAOHANSEYNIONO
CMLDEPIULOIUTIPLCRNNOHAPLSANOASAWSEUECTLSETREPIULOIUTHNLTHLXLGAFANOPNXYAVAPGITIWSEUECTLSETREPIULOIUI
LQNFATEXLGAFANOIPAGAFAPAGAFWSEUECTLSEAODTREPIULOIUNHRLTANNOPAOAULPLIPAWSEUECTLSEATTREHAFETNFETRE
SEVESEAMHIXEOWIPIEULONOCRLYIPLSEYNIOAODPGLHRLUISEJAFWMETREAOACNEOTTIFBHIUFIITHRNOAVANOTRECNTGUAHARN
PGITIWSEUECTLSEAODTREAOACNEOTTIFBHIUTIFNPAYANOTRECNTGIUPAGAFAAFIOTITRESHTREMASYEHTIOENHTRETITPLTIFAAOC
NEOTTIFBTRETITAMMEOYTRIUTREFILODNHABILTTVIRLODSEDENYRTGFETESHVRNCRNHFISETRAOTRSEETNFEHMASYESTRAOTREMA
SYEHTFILODTIFBIUABILTENYRTGFETESHNOTREMATESWESNIDIUTREGAGINWESNIDAODNTHASEAAODCAWACNTGASEUASYSEATESTR
AOTREMATTESAMTRILYRTREFILODTIFBHVNTAUSIOTAODBACPCNSCLMASHRAWVESEAMSEADGHGATTESEDTRSILYRILTQAWAODLSN
OYTREGAGINWESNIDTREEASMGUISFHIUTREXEOWIPIEULOVESEFIHTMGKESTNCAMCAKEHTGMEHTIOECRAFVBESHVNRBAFBIHRAW
DVIIDEOCIUUNOHVRNCRVESEDNUVESEOTUSIFTREFILODTIFBHDLSONYTTREGAGINWESNIDFISEIKESEASMGXEOWIPIEULORADCIFF
IOUEATLSEHHL CRAHAMIVESAODVNDUSIOTWASTCWFASEDTITREBACPCNSCLMASWASTAODTRECIFBNOATNIOADWIHNTNIOIUYSA
KEYIIDHVESETREHAFENOHNDRECIUUNOVESEQADEAODFNSSISHVRNMEILTHNDETRECIUUNOVESEAMASYEOLFBESIUTSNAOYLMASE
DYEDKNNOEBEAHTFNSSISHNSIOVEAWIOHAYSNCLMTLSAMAODUNHRNOYTIIMHAODHIIIOSEYASDNOYTRNHTANNCRNSIHRNSANHRNBEMN
EKEDTRATATTRATTNFEKASNILHUISCEHNOPNOIMEDBGTREGAFATANPNODYIFAODTREHETINOMADHEAUISEFEDAOAMNNOCEAOIDBTA
NOEDNSIOAODKASNILHCLMTLSAMSEMNCUSIFPISEABGDEUEATNOYUISCEHHL CRAHOLPIPLAODTRENTIPNOYDIFHVRNCRCIOTSIMME
DTREYEOPANHEATOTREITRESAODGIHRNSIPIODIAMHIFEOTNIOEDTRATFIHTIUTREXEOWIPIEULOBBLNMTIOMATMAODVESEVEMMI
SYAONXEDAODTREFILODHVESEAMHIBLNM TMAGESBGMAGESAUTESCIOHIMNDATNIOCIOHNDENSOYTREHMIWEAODASSAOYFEFOTIUTRE
FILODH CNSCLMASFILODHVESECIOHTSLCTEDNOADDNTNIOBSIOXEFNSSISHUSIFFANOMAODCRNOAAODTREVAQNODEOUSIFTREHAFEW
ESNIDNODNCATETRATTRESEVAHAOEJCRAOYEBETVEEOTRETVICILOTSNEHATTRATTNFETRESEUISENTNHHWECLMATEDTRATTREEASM
GAWWEASAOCEIUTREXEOWIPIEULOFAGRAKEBEEONUOMLEOCEDBGCNKNMEOYNOEESNOYTECROIMIYGUSIFFANOMAODCRNOAAODTREW
IMNTNCAMAODNDEIMINYCAMNOUMLEOCENHSEUMECTEDNOTREQINOTCIOHTSLCTNIOIUTREXEOWIPIEULOBGKASNILHAMNNOCEFEF
BESHAWHSIIUIUTREAMNNOCEATTREHAFETNFETRESEASEAMHITIFBHUNKEULSAODTRSEEIUTREPAFEOLFONOTREUSIOTIUTRERIL

```

GLAOTIFBNOEAHTESQAWAODLSNOYTREHAFEWESNIDAHNRNBLTANPIULOAFIOYTREFTIFBUNKENHTREEASMNEHTPIULONOEAHTESO
 QAWAOHRNHTISGNODNCATNOYAWIHHNBMECIOECTNIOBETVEOTREPAXLHAWSIKNOCEAODTREGAFATANPNODYIFATTRATTNFENOTRE
 UILSTRCEOTLSGRIPLSNPLSEYNIOPAOTISEYNIOTIPANSEYNI OAODTIRIPLSEYNI OVESEFANOMGCIFWIHEDIUSIOTAODSEASTIFBH
 VRNCRYSADLAMMBECAFEXEOWIPIEULOTVITITRSEEYEOESATNIOHMATESSEYASDNOYTRNHGAFATITAPESLWINOTEDILTTRATAMTR
 ILYRTREPIQNPNAODONRIOHRIPNFEOTNIOEDTRATTREUILSYEOESAMHAODQAWAOEHETAPEXLOAODITRESHRADFESYEDEAHTQAWAONO
 TITRETESNTISGUTREQAWAOEHEFIOASCRGTRSILYRFLMTNWMEEJWEDNTNIOHREBEMNEKEDTRATNTVAHACTLAMMGLOLOIPLONMIC
 ATEDNOTREOIBNWMANOTRATUISFEDAOAMMNAOCENOEHTQAWAOFANOMGCIOHNHTNOYIUSEASYSAKEHNOTREWAHTAODTRATAUTESTRE
 DEATRIURNFPITREGMIHTTREBATTMEVNTRTREGAFATANPNODYIFISOEYITNATEDWEACELODESTREMEADESHRNWIUTREGAFATIPNOY
 HRNWNOTREODTREAMMNAOCEIUTREDIYHMAKEPNOYDIFAODTREAMMNAOCEIUTREKNMRISHEPNOYDIFESYEDNOTIOEUISFNOYTRE
 QAWAOEHEFIOASCRGTREOTREMEADESHIUEAHTAODVEHTQAWAOVESEDNHTNOYLNHREDBGTREXEOWIPIEULOAOOTREUSIOTAODSEAST
 IFBHUSIFTREEODIUTREUILSTRCEOTLSGTITREUNTRCEOTLSGMASYEXEOWIPIEULOSEHEFBMNOYPIULONOFILOTTANDABEYAOTIA
 WWEASNOWSEHEOTDAGYLOFAWSEUECTLSEMASYEXEOWIPIEULORAKEAMHIBEEOBLNMTNOWMACEHLCRAHGAF AOAHRNWSEUECTLSENB
 ASAPNWSEUECTLSEAODCRNBWSEUECTLSEAUTESTREFNDUILSTRCEOTLSGTREDAVATIFBHSYNNOAMMGCEOTESEDASILODTREGAFAT
 IPULONOTREHILTREAHITUTREOASABAHNOAODTRESGLBEOPULYOYADLAMMGHRNUTEDTITREDANHEOPIULONOTREOISTRTREMASYE
 XEOWIPIEULOLOIUDAVAVAHTREULSLNCRNPULOCMLHTESAODFXLTIFBHTRECEOTESIUYSAKNTGIUTREYSEATPNODYTIFBHRNUTEDA
 YANONOTREUNTRCEOTLSGTITREAOCEOTCNTGTIFBYSILWAOOTRERLODSEDTIOYLEDNBSDTIFBYSILWNOTREIHAPAWMANOSEYASDN
 OYTRESEAHIOHUISTREFLMTNWMETSAOHUESHIUTREHEMASYEXEOWIPIEULOBAHEDIOSEHEASCRITOTREPLHRNQNAODONRIOHRLQNTR
 EQAWAOEHEACADEFNCCIFFLONTGRAHWSIWIHEDKNEVHHL CRAHTRETREISGIUDGOAHTGAMTESOATNIOAODTRETREISGIEZLEHTSAOT
 SNBEHCIOZLESNOYDGOAHTNEHTIEJWMANOIOTREITRESRAODRNSIHEPAXLWSIWIHEDTRATTREHTSLCTLSEHIUTREHAHAPNP IQHNRN
 XLPAODFAPNOIAOCNEOTTIFBYSILWHVESEHNFNMASAOUSIFTREMAUEILSTRCEOTLSGTITREHECIODRAMUIUTREUNTRCEOTLSGT
 REREADHIUTREUILSAOCNEOTTIFBYSILWHQINOTMGWMAGEDASIMENOTREGAFATIPNOYHRNWAUTESDEATRREGDENUNEDAHTREYLASD
 NAODENTGIUTREGAFATISIGAMWIVESAODAMHIWSIWIHEDTRATTRECIOTSLCTNIOIUTREMASYEXEOWIPIEULORADADETESSEOTEUU
 ECTIOKASNILHSEYNIOHATTREHAFETNFENOTREHECIODRAMUIUTREUNTRCEOTLSGOIMASYEHCAEXEOWIPIEULOVESECIOTSLCT
 EDNOASEAHHL CRAHPNBNWSIKNOCELWESFAIGEAO DRGLYAWSIKNOCEAODTREHCAMEIUTREXEOWIPIEULONNOYIPNHRNCRNDIYSADLA
 MMGDECSEAHEDACCISDNOYTTIRENOHCSNWTNIOHQNTRENSIOHVISDLOEASTREDUSIFTRENOAVAFILOTANOPIULOAOOTREMASYE
 HVISDLOEASTREDUSIFTREETAULOAGAFAPIULOEFWESISGLSGAPLATTRATTNFSEYASDEDRNFHEMUAHTREPNOYIUTREVISMADOARNH
 ALTRISNTGVAHSEUMECTEDNOTREUACTTRATIOMGRECIOTNOEDTIBLNDMASYEXEOWIPIEULONOTREQAWAOEHEASCRNWEMAYINOTR
 EHNJTRCEOTLSGTRERISNXIOTAMCAKEHTGMEHTIOECRAFBSIUTREPNOANHGHTEFBYEAOTIBEWIWLMSXEDNOXEOWIPIEULOATTR
 EHAFETNFETREGBEYAOTIAWWEASNOASEAHHL CRAHHTHLSIKNOCEIVASNWSIKNOCELQNWSIKNOCEAODEKEONOPAOITSEYNIORVNC
 RVESECIOTNDESEDAHBAHEHUISTREEHTABMNRHFEOTIUTREEFWESISPENTANAFIOYTREFLEOIRADAWASTNCLMASMGMASYEO LFBESI
 XEOWIPIEULOCIFWASEDTIOMGTRNSTGONOENOPIXLPEWSIKNOCEDLSNOYTREHAFEWESNIDLEOIAMIOERADONOTGHEKEOVRNMENO
 PAOTITRESEVESETVIRLODSEDHNTJTEOAHBITRPNIOAODINXRAOYDECSEAHEDIOGTREPAOTISEYNIOCIOTNOEDTICIOHTSLCTMAS
 YEXEOWIPIEULOTRSILYRILTTREHNJTRCEOTLSGNODNCATNOYTRATTREGAFATIFIOASCRGATTRATTNFSEMNEDREAKNMGIOTREAH
 TESOPNOYDIFNOBITRECIOIFNCAODFMNTASGAHWECTHUSIFTREEODIUTREHNJTRCEOTLSGTITREBEYNOONNOYIUTREHEKEOTRCEOTL
 SGMASYEHZLASEAODCNSCLMASTIFBHSEWMACEDTREUSIOTAODBECAFETREFANOHTSEAFNOTREFNDMEIUTREHEKEOTRCEOTLSGTREG
 EKIMKEDNOTIICTAYIOAMTIFBHSEYASDNOYTRESEAHIOUISTRECEHATNIOIUCIOHTSLCTNIOIUXEOWIPIEULOTANNCRNSIHRNSAN
 HRNBEMNEKEHTRATNTNHSEMATEDTITRESEUISFHIUCSIVOWSNOCEHRTIPLAODHIYA OILFAPIAHVEMMAHTREEHTABMNRHFEOTIUTRE
 OATNIOAMFAOLUACTLSNOYHGHTFVRNMETREUNOAMXEOWIPIEULOASEBEMNEKEDTIBEAHAFILOTANOPIULOADSIOYFACRNCMAN
 FHTRATTREPIULOVESEBLNMTNOTREUNSHTRAMUIUTREHEKEOTRCEOTLSG

4. Cocokkan melalui informasi kecocokan bigram dan trigram, lalu lanjutkan dengan iterasi yang sama: setiap kali sebuah bigram/trigram/kata umum Bahasa Inggris mulai terlihat pada hasil dekripsi semantara, huruf *cipher* yang bersangkutan diganti agar sesuai. Tiga huruf – R, Q, U – sudah benar sejak tebakan awal (langkah 1) dan tidak pernah diubah lagi. Untuk sisanya, dapat dilihat pada tabel di bawah yang disusun berdasarkan *log* dari *src/21/log/plain/* secara kronologis.

Ganti	Konteks (sebelum → sesudah)	Petunjuk
P→H	AUTESTREFNDONOETEEOTR → AUTESTHEFNDONOETEEOTH	delapan karakter pertama UMRQAR PQ → AUTESTHE, trigram THE
I→N	UTESTHEFNDONOETEEOTHCEOTL UTESTHEFNDNNNTEENTHCENTL →	mengarah ke NINETEENTH
C→I	AUTESTHEFNDNNNTEENTHCEN AUTESTHEFIDNINETEENTHCEN →	kata NINETEENTH terbentuk
A→R	AUTESTHEFIDNINETEEN → AUTERTHEFIDNINETEEN	bigram ER
Z→U	TEENTHCENTLSGMASYEHCAMEXE TEENTHCENTURYMARYEHCAMEXE →	mengarah ke CENTURY
B→Y	ENTHCENTLSGMASYEHCAMEXENW ENTHCENTURYMARYEHCAMEXENW →	kata CENTURY lengkap
M→F	AUTERTHEFIDNINET → AFTERTHEFIDNINET	kata umum AFTER THE
N→M	AFTERTHEFIDNINETEENTHCE AFTERTHEMIDNINETEENTHCE →	kata MID (AFTER THE MID-...)
D→O	EHCAMEXENPIPIENFUNBEYANTI EHCAMEXENPOPOENFUNBEYANTO →	mengarah ke ZENPOKOENFUN
T→S	NTURYMARSEHCAMEXENPOPOENF NTURYMARYESCAMEXENPOPOENF →	bigram ES
F→K	SCAPEXENPOPOENFUNBEYANTOA SCAPEXENPOKOENFUNBEYANTOA →	mengarah ke ZENPOKOEN
S→G	CENTURYPARYESCAPEXENPOKOE CENTURYPARGESCAPEXENPOKOE →	kata LARGE
O→P	GESCAPEXENLOKOENFUNBEGANT GESCAPEXENPOKOENFUNBEGANT →	kata ZENPOKOEN
X→L	NTHCENTURYPARGESCAPEXENPO NTHCENTURYLARGESCALEXENPO →	kata LARGE SCALE
W→W	TOAPPEARINGESTERNQAPANEJA TOAPPEARINWESTERNJAPANEXA →	kata IN WESTERN
G→J	RINGESTERNQAPANEJAMPLESOF RINWESTERNJAPANEXAMPLESOF →	kata JAPAN
J→X	TERNQAPANEJAMPLESOFKANSAI TERNJAPANEXAMPLESOFKANSAI →	kata EXAMPLES
Y→Z	LARGESCALEXENPOKOENFUNBEG LARGESCALEZENPOKOENFUNBEG →	istilah ZENPOKOENFUN terbentuk
E→V	WEREMOSTLYKERTICALCAKESTY WEREMOSTLYVERTICALCAVESTY →	kata VERTICAL CAVE
L→Q	LARGESZUAREANDCIR → LARGESQUAREANDCIR	kata SQUARE

Artefak

Skrip *solver* yang digunakan untuk menghitung frekuensi monogram/bigram/trigram, menyusun tebakan awal, menerapkan pemetaan hasil analisis, serta mencatat setiap hasil dekripsi ke log/plain/ dan metadatanya (pemetaan & frekuensi n-gram) ke log/metadata/. Modul ini bergantung pada `src/frequencies.py` yang berisi tabel referensi frekuensi Bahasa Inggris (dipakai bersama untuk seluruh soal substitusi/Vigenère pada laporan ini, sehingga isinya tidak diduplikasi di sini).

```

1 import json
2 import os
3 import sys
4 from datetime import datetime
5 from typing import Dict, List
6
7 sys.path.insert(0, os.path.dirname(os.path.dirname(os.path.abspath(__file__))))
8 from frequencies import MONOGRAM, MONOGRAM_ORDER
9
10 CORRECTIONS = [
11     ("R", "T"), ("P", "H"), ("Q", "E"), ("I", "N"), ("C", "I"),
12     ("U", "A"), ("Z", "U"), ("A", "R"), ("B", "Y"), ("M", "F"),
13     ("O", "P"), ("W", "W"), ("N", "M"), ("D", "O"), ("T", "S"),
14     ("X", "L"), ("G", "J"), ("F", "K"), ("S", "G"), ("J", "X"),
15     ("Y", "Z"), ("E", "V"), ("L", "Q"),
16 ]
17

```

```

18 class Solver:
19     def count(self, string: str) -> Dict:
20         data = {}
21         for c in string:
22             data[c] = data.get(c, 0) + 1
23         return dict(sorted(data.items(), key=lambda item: item[1], reverse=True))
24
25     def sort(self, string: str) -> List:
26         return list(self.count(string).keys())
27
28     def freq_ngram(self, string: str, n: int = 1) -> Dict:
29         freq_data = {}
30         total_ngrams = len(string) - n + 1
31         for i in range(total_ngrams):
32             ngram = string[i:i+n]
33             freq_data[ngram] = (freq_data.get(ngram, 0) + 1)
34         for ngram in freq_data:
35             freq_data[ngram] = (freq_data[ngram] / total_ngrams * 100)
36         return dict(sorted(freq_data.items(), key=lambda item: item[1], reverse=
37 True))
38
39     def solve(self, string: str) -> Dict:
40         cipher_order = [c for c in self.sort(string) if c.isalpha()]
41         mapping = {}
42         for cipher_letter, plain_letter in zip(cipher_order, MONOGRAM_ORDER):
43             mapping[cipher_letter] = plain_letter
44         return mapping
45
46     def replace_char(self, mapping: Dict, char0: str, char1: str) -> Dict:
47         mapping[char0] = char1
48         return mapping
49
50     def decrypt(self, string: str, mapping: Dict) -> str:
51         return "".join(mapping.get(c, c.lower()) for c in string)
52
53     def table(self, string: str) -> str:
54         cipher_freq = self.freq_ngram(string)
55         rows = ["rank cipher cipher% -> plain plain%"]
56         for i, cipher_letter in enumerate(cipher_freq):
57             if not cipher_letter.isalpha():
58                 continue
59             plain_letter = MONOGRAM_ORDER[i] if i < len(MONOGRAM_ORDER) else "-"
60             plain_pct = MONOGRAM.get(plain_letter, 0.0)
61             rows.append(
62                 f"{i+1:>4} {cipher_letter:^6} {cipher_freq[cipher_letter]:>6.2f}
63                 -> "
64                 f"{plain_letter:^5} {plain_pct:>5.2f}"
65             )
66         return "\n".join(rows)
67
68     def load_ciphertext(self, base_dir: str) -> str:
69         with open(os.path.join(base_dir, "s.txt")) as f:
70             return f.read().strip()
71
72     def write_plain(self, base_dir: str, timestamp: str, decrypted: str) -> str:
73         path = os.path.join(base_dir, "log", "plain", timestamp + ".txt")
74         os.makedirs(os.path.dirname(path), exist_ok=True)
75         with open(path, "w") as f:
76             f.write(decrypted)

```

```

75     return path
76
77 def write_metadata(base_dir: str, timestamp: str, mapping: Dict, freq: Dict) -> str
78 :
79     path = os.path.join(base_dir, "log", "metadata", timestamp + ".json")
80     os.makedirs(os.path.dirname(path), exist_ok=True)
81     payload = {
82         "timestamp": datetime.now().isoformat(timespec="seconds"),
83         "mapping": mapping,
84         "freq_ngram": {str(n): freq[n] for n in freq},
85     }
86     with open(path, "w") as f:
87         json.dump(payload, f, indent=2)
88     return path
89
90 if __name__ == "__main__":
91     base_dir = os.path.dirname(os.path.abspath(__file__))
92     content = load_ciphertext(base_dir)
93
94     solver = Solver()
95     mapping = solver.solve(content)
96     for cipher_letter, plain_letter in CORRECTIONS:
97         mapping = solver.replace_char(mapping, cipher_letter, plain_letter)
98
99     decrypted = solver.decrypt(content, mapping)
100     freq = {n: solver.freq_ngram(content, n) for n in (1, 2, 3)}
101
102     timestamp = datetime.now().strftime("%Y-%m-%d_%H%M%S")
103     plain_path = write_plain(base_dir, timestamp, decrypted)
104     metadata_path = write_metadata(base_dir, timestamp, mapping, freq)
105
106     print("mapping (cipher -> plain):")
107     print("".join(sorted(mapping)))
108     print("".join(mapping[c] for c in sorted(mapping)))
109     print()
110     print("hasil dekripsi:")
111     print(decrypted)
112     print()
113     print(f"[tersimpan ke {os.path.relpath(plain_path)} dan {os.path.relpath(
114         metadata_path)}]")

```

src/21/solver.py

Plaintext Hasil Dekripsi

AFTER THE MIDNINETEENTH CENTURY LARGE SCALE ZENPO KOEN FUN BEGAN TO APPEAR IN WESTERN JAPAN. EXAMPLES OF KANSAI REGION INCLUDE KOFUN OF TOKUCHI IN SAKURAI NARA PREFECTURE, THE KOFUN OF TSUITSUZYAMA IN KIZUGAWA KYOTO PREFECTURE, THE KOFUN OF FUJIMATEZUYAMA IN OKAYAMA OKAYAMA PREFECTURE AND THE KOFUN OF ISHIBUTAI IN KANFUKU OKAPREFECTURE AT THE SAME TIME THERE WERE ALSO ZENPO KOEN FUN IN CHUGOKU REGION AND KYUSHU. FOR EXAMPLE THE ANCIENT TOMBS OF MOTOSHI IN NAWA IN THE CITY OF ASAHIKYOTO PREFECTURE AND THE ANCIENT TOMBS OF TOMIKAGA IN THE CITY OF OKAYAMA AMONG OTHERS. THE LARGEST ONE IS THE TOKUTOMA ANCIENT TOMB. THE TOTAL LENGTH OF THE MOUND IS ABOUT TWO HUNDRED EIGHTY METERS, WHICH IS MORE THAN THREE TIMES LARGER THAN THE LARGEST MOUND TOMB OF ABOUT EIGHTY METERS IN THE LATER PERIOD OF THE YAYOI PERIOD AND ITS AREA AND CAPACITY ARE FAR GREATER THAN THE LATTER. ALTHOUGH THE MOUND TOMBS WITH A FRONT AND BACK CIRCULAR SHAPE WERE ALREADY SCATTERED THROUGHOUT JAPAN DURING THE YAYOI PERIOD, THE EARLY FORMS OF THE ZENPO KOEN FUN WERE MOSTLY VERTICAL CAVE STYLE STONE CHAMBERS WITH BAMBOO SHAPED WOODEN COFFINS WHICH WERE DIFFERENT FROM THE MOUND TOMBS DURING THE YAYOI PERIOD. MORE OVER EARLY ZENPO KOEN FUN HAD COMMON FEATURES SUCH AS A LOWER AND WIDER FRONT PART COMPARED TO THE BACK CIRCULAR PART AND THE COMBINATION AND POSITION OF GRAVE GOODS WERE THE SAME INSIDE THE COFFIN. THERE WERE JADE AND MIRRORS WHILE

EOUTSIDETHECOFFINWEREALARGENUMBEROFTRIANGULAREGDEDIVINEBEASTMIRRORSIRONWEAPONSAGRICULTURALANDFISHINGTOOLSANDSOONREGARDINGTHISTAIICHIROSHIRAIISHIBELIEVEDTHATATTHATTIMEVARIOUSFORCESINKINOLEDBYTHEYAMATAIKINGDOMANDTHESETOINLANDSEAFORMEDANALLIANCEANDOBTAINEDIRONANDVARIOUSCULTURALRELICSFROMKOREABYDEFEATINGFORCESSUCHASNOKUKUANDTHEITOKINGDOMSWHICHCONTROLLEDTHEGENKAISEAONTHEOTHERHANDYOSHIROKONDOALSOMENTIONEDTHATMOSTOFTHEZENPOKOENFUNBUILT ON FLATLAND WERE WELL ORGANIZED AND THE MOUNDS WERE ALSO BUILT LAYER BY LAYER AFTER CONSOLIDATION CONSIDERING THE SLOPE AND ARRANGEMENT OF THE MOUNDS CIRCULAR MOUNDS WERE CONSTRUCTED IN ADDITION BRONZE MIRRORS FROM MAINLAND CHINA AND THE WAJINDEN FROM THE SAME PERIOD INDICATE THAT THERE WAS AN EXCHANGE BETWEEN THE TWO COUNTRIES AT THAT TIME THEREFORE IT IS SPECULATED THAT THE EARLY APPEARANCE OF THE ZENPOKOEN FUN MAY HAVE BEEN INFLUENCED BY CIVIL ENGINEERING TECHNOLOGY FROM MAINLAND CHINA AND THE POLITICAL AND IDEOLOGICAL INFLUENCE IS REFLECTED IN THE JOINT CONSTRUCTION OF THE ZENPOKOEN FUN BY VARIOUS ALLIANCE MEMBERS AS PROOF OF THE ALLIANCE AT THE SAME TIME THERE ARE ALSO TOMBS FIVE FOUR AND THREE OF THE KAMENMUN IN THE FRONT OF THE HOUYUAN TOMB IN EASTERN JAPAN DURING THE SAME PERIOD AS ISHIBUTAI KOFUN AMONG THE TOMB FIVE IS THE EARLIEST KOFUN IN EASTERN JAPAN HIS TORY INDICATING A POSSIBLE CONNECTION BETWEEN THE KAZUS A PROVINCE AND THE YAMATAI KINGDOM AT THAT TIME IN THE FOURTH CENTURY HOKURIKU REGION KANTO REGION TOKAI REGION AND TOHOKU REGION WERE MAINLY COMPOSED OF FRONT AND REAR TOMBS WHICH GRADUALLY BECAME ZENPOKOEN FUN TWO TO THREE GENERATIONS LATER REGARDING THIS YAMATO TAKERU POINTED OUT THAT ALTHOUGH THE KOJIKI AND NIHON SHOKI MENTIONED THAT THE FOUR GENERALS AND JAPANESE TAKEZUN AND OTHERS HAD MERGED EAST JAPAN INTO THE TERRITORY OF THE JAPANESE MONARCHY THROUGH MULTIPLE EXPEDITIONS SHE BELIEVED THAT IT WAS ACTUALLY KUNUNOKUNI LOCATED IN THE NOBI PLAIN THAT FORMED AN ALLIANCE IN EAST JAPAN MAINLY CONSISTING OF REAR GRAVES IN THE PAST AND THAT AFTER THE DEATH OF HIMIKO THEY LOST THE BATTLE WITH THE YAMATAI KINGDOM OR NEGOTIATED PEACE UNDER THE LEADERSHIP OF THE YAMATO KINGSHIP IN THE END THE ALLIANCE OF THE DOGSLAVE KINGDOM AND THE ALLIANCE OF THE EVIL HORSE KINGDOM MERGED INTO ONE FORMING THE JAPANESE MONARCHY THEN THE LEADERS OF EAST AND WEST JAPAN WERE DISTINGUISHED BY THE ZENPOKOEN FUN AND THE FRONT AND REAR TOMBS FROM THE END OF THE FOURTH CENTURY TO THE FIFTH CENTURY LARGE ZENPOKOEN FUN RESEMBLING KOFUN IN MOUNT TAIDABEGAN TO APPEAR IN PRESENT DAY GUNMA PREFECTURE LARGE ZENPOKOEN FUN HAVE ALSO BEEN BUILT IN PLACES SUCH AS YAMANASHI PREFECTURE IBARAKI PREFECTURE AND CHIBI PREFECTURE AFTER THE MID FOURTH CENTURY THE DAWA TOMBS ORIGINALLY CENTERED AROUND THE YAMATO KOFUN IN THE SOUTHEAST OF THE NARABASIN AND THE YUBEN KOFUN GRADUALLY SHIFTED TO THE DAISEN KOFUN IN THE NORTH THE LARGE ZENPOKOEN FUN OF DAWA WASTHE FURUICHI KOFUN CLUSTER AND MOZUTO MBSTHE CENTER OF GRAVITY OF THE GREAT KING TOMBS SHIFTED AGAIN IN THE FIFTH CENTURY TO THE ANCIENT CITY TOMB GROUP AND THE HUNDRED TONGUE DBIRD TOMB GROUP IN THE OSAKA PLAIN REGARDING THE REASONS FOR THE MULTIPLE TRANSFERS OF THESE LARGE ZENPOKOEN FUN BASED ON RESEARCH ON THE KUSHIJI AND NIHON SHUJI THE JAPANESE ACADEMIC COMMUNITY HAS PROPOSED VIEWSSUCH AS THE THEORY OF DYNASTY ALTERNATION AND THE THEORY OF QUESTER AN TRIBES CONQUERING DYNASTIES TO EXPLAIN ON THE OTHER HAND HIROSE KAZUO PROPOSED THAT THE STRUCTURES OF THE SAKI KOJISHI ZUKI AND MAKINO ANCIENT TOMB GROUPS WERE SIMILAR AND FROM THE LATE FOURTH CENTURY TO THE SECOND HALF OF THE FIFTH CENTURY THE HEADS OF THE FOUR ANCIENT TOMB GROUPS JOINTLY PLAYED A ROLE IN THE YAMATO KINGSHIP AFTER DEATH THEY DEIFIED AS THE GUARDIAN DEITY OF THE YAMATO ROYAL POWER AND ALSO PROPOSED THAT THE CONSTRUCTION OF THE LARGE ZENPOKOEN FUN HAD A DETERRENT EFFECT ON VARIOUS REGIONS AT THE SAME TIME IN THE SECOND HALF OF THE FIFTH CENTURY NO LARGE SCALE ZENPOKOEN FUN WERE CONSTRUCTED IN AREAS SUCH AS KIBI PROVINCE UPPER MAOE AND HYUGA PROVINCE AND THE SCALE OF THE ZENPOKOEN FUN IN GOKISHI CHIDO GRADUALLY DECREASED ACCORDING TO THE INSCRIPTIONS ON JIJITHE IRON SWORD UNEARTHED FROM THE INAWAMOUNTAIN KOFUN AND THE LARGE SWORD UNEARTHED FROM THE ETAFUNAYAMA KOFUN EMPEROR YURYAKU AT THAT TIME REGARDED HIMSELF AS THE KING OF THE WORLD AND HIS AUTHORITY WAS REFLECTED IN THE FACT THAT ONLY HE CONTINUED TO BUILD LARGE ZENPOKOEN FUN IN THE JAPANESE ARCHIPELAGO IN THE SIXTH CENTURY THE HORIZONTAL CAVE STYLE STONE CHAMBER OF THE KINAI SYSTEM BEGAN TO BE POPULARIZED IN ZENPOKOEN FUN AT THE SAME TIME THEY BEGAN TO APPEAR IN AREAS SUCH AS SETSU PROVINCE OWARI PROVINCE UJIPROVINCE AND EVEN IN KANTO REGION WHICH WERE CONSIDERED AS BASES FOR THE ESTABLISHMENT OF THE EMPEROR KEITAI AMONG THEM UENO HAD A PARTICULARLY LARGE NUMBER OF ZENPOKOEN FUN COMPARED TO ONLY THIRTY NINE IN KOZUKE PROVINCE DURING THE SAME PERIOD UENO ALONE HAD NINETY SEVEN WHILE IN KANTO THERE WERE TWO HUNDRED SIXTEEN AS BOTH KINO AND OIZHANG DECREASED ONLY THE KANTO REGION CONTINUED TO CONSTRUCT LARGE ZENPOKOEN FUN THROUGHOUT THE SIXTH CENTURY INDICATING THAT THE YAMATO MONARCHY AT THAT TIME RELIED HEAVILY ON THE EASTERN KINGDOM IN BOTH ECONOMIC AND MILITARY ASPECTS FROM THE END OF THE SIXTH CENTURY TO THE BEGINNING OF THE SEVENTH CENTURY LARGE SQUARE AND CIRCULAR TOMBS REPLACED THE FRONT AND BECAME THE MAINSTREAM IN THE MIDDLE OF THE SEVENTH CENTURY THEY EVOLVED INTO OCTAGONAL TOMBS REGARDING THE REASON FOR THE CESSATION OF CONSTRUCTION OF ZENPOKOEN FUN TAIICHIROSHIRAIISHIBELIEVES THAT IT IS RELATED TO THE REFORMS OF CROWN PRINCESHOTO KU AND SOGANO UMAKO AS WELL AS THE ESTABLISHMENT OF THE NATIONAL MANUFACTURING SYSTEM WHILE THE FINAL ZENPOKOEN FUN ARE BELIEVED TO BE AS A MOUNTAIN KOFUN AND RONGMACHI CLAIMS THAT THE KOFUN WERE BUILT IN THE FIRST HALF OF THE SEVENTH CENTURY

Kunci

cipher	ABCDEFGHIJKLMNOPQRSTUVWXYZ
plain	RYIOVKJBXNDQFMPHETGSACWLZU

Referensi

Tabel referensi frekuensi monogram, bigram, dan trigram Bahasa Inggris yang digunakan pada langkah-langkah di atas mengacu pada [3, 4].

2.2 Horizon Chartreuse

Deskripsi Soal

Di lembar catatan riset berikutnya, Marisa menemukan laporan mengenai fenomena aneh yang menimpa langit malam. Seseorang telah mencuri bulan purnama yang asli dan menggantinya dengan purnama palsu yang redup. Laporan yang memuat kronologi hilangnya bagian dari langit tersebut kini tidak dapat dibaca karena seluruh teksnya dienkripsi menggunakan **Vigenère Cipher**.

Manfaatkan metode **Kasiski Examination** untuk mengidentifikasi perulangan pola karakter pada *ciphertext* dalam memperkirakan panjang kuncinya. Setelah panjang kunci diperoleh, terapkan analisis frekuensi bahasa Inggris pada tiap sub-teks untuk merekonstruksi kunci dan memulihkan seluruh isi naskah.

Berkas lampiran:

- Attachment: https://drive.google.com/file/d/1LyCID57V3ckzlvQ9Wnj-mGg_YMH0P50e/view?usp=sharing
- md5sum: 6b933809bbedb3c5c0f1fd9e2b1076cd
- sha256sum: d6ecf3775c56593f833db5c62e1447f2668b41a310562e7a9ccd6a1486d5f641

Berkas *Ciphertext*

```
MHPQVEQYKKUJLRXGWGJEIFHWFVIOBLCLRXPZODJYGVAYTZYHDYBYIISWXINTMLRXPQIHWFJIEFWQFHMURXEVONALUEHGFBTOUXBZRDN
SOUZCGDGFZNUYMUOOIKVUKMLHUXQRAAGUWSBXORVIRPDYGLQVFIIXLIYFIWHTGZNMHDPWEWSHAOZDRGRZYXWGLZVYMGZMQVVIKBCNVNYV
BEJUWALMYEQEQAQMOMPVJRVGBKPFJRMJUIHSXHVJLKGPKYCFDRFYLSHGMVRRCBKGAXXNWXYRMJNXISJIEFWQFHCVEGCMGYTHPHVRQGG
AXMGKIWRMRPGABIMKFPZMIEBEEQIPGXBIKZWYWAYEHXYNBXLCYPWMMVAOAHFLXVZAZOTNYRSBNZOHOTKXYRGSZBGEIWEWSAAAYBXX
VEEUYEXUDZVFWSLAIJISLALZOXCEZEPVVZVMBVVLFAKZMLNQKVZIIIDHRVFFBSYAAUGPEBRVUZXFHFQRTIFPGYFRVFCIOHVVHGKNVOYH
HORBKUIZDTMBQGVNBKSPVVMFAVUDLYEYIJGPKYHFRJJJCQXPMONOWKBZGNXMCVZGAIHGFVYINRTRPGMBPIKUQTNMNBWYJRLZVWCFSPLR
PATTHFKEGRAYOXXFDCBPKYLBVNMXNUOAAACFLWKUMNVFYBICLXIXPRUXXQFGPKFHOXDMFSJUBGXNUMVFMBLKVMQGVOMIVFCAJENNZKVYUF
OMXUBGIGIEPECVBEQARQWFGUZAYUDWEGJKLGOOIKBARLXJHPCRDKUWOLRXGPKKTSKGIVAMSFPUBFIKGWHLRYRQFLGANLPUFFIIGIO
UMBNMWMKZCYAAUIHXRXMTZHGRRRCWCLKZHOXFCRSMNBWSWSANLWIRVKTMTZVNBIXVAPUDXPRUEEQZDBFYEIMRZEIHNUHVJBKMHUHV
KBBGRXUADGKGVTHEIAHLVLQIHGLBSIYZOUTHQOIKUMXOTHQOIKUQYWKIOOIDFWYOMRWSLGNUYMBRVLIVVKHGIGKIIOCOSWCAJFPPIZLW
UGWLVRMLHZTHRJBSEVMBRUIJNPATTHGKEKFPKRGIJVYXRZPMNFEIRNXLJRUFVAENVTFJDCJUIVWXHFWSSRJUYXXFRWYRTRNEUQOCLA
LKYMUXHYRBGZDISUIJBTBGAJKEKRDKYMFBXFCRBNLYVVMKFUKSEMBPMEBCYPMCFVEZQBNNHNUHJFEMYAXUGVLLZITZAOZDRJHAAHEFL
PEZABGPGUFDJVQYATHPHJIBUOABNFDXDBAVOXLRFSEFBGUMFLEVZZACPMBNQYERIXAAFLHZZYBNLYIEHWKBNSHZCPDTCNKKDAYEHECYBN
LXPVOMETMTZHELRRRGXHEFLJEKUMXZPCGKMEGPGAYIEHWKFUGSEBHPESQMBKYFDVVTIIZOXLRGMEGWZFYUOFLVTJPGAUXQRANONNLRVW
DNTRLKNUDRYHUGULNUHVRVHVVPWOSIRLVBIRJWIVEIRPVYZDVXNBXVBXVVVVNLOUZVBROJNUOKLNNPSLABGPGISGSCYANVPWNQLLZITZI
IFVMSYGTVMVRDAREMUMMBVLLTMJPLUFWIIQCTUHCSSQKUQTNBMQRRVJMCVGNHESYMZVXHWCKUIZZAYVVRKGPKARJRRJGRZYVGQURKVG
AUBMISWIEOCZUHIAHICFMCHLCAWIIIRAZLWCAWLVCZUIEYZVSJUMYLMHIXFVVBLLNVJEKRNUYAYEVICSWXYTNHUVJUMZYBYQWSZQWTAECX
HXYVAZOXSEHYJRLZVMBVVOZALUMMBVQKJBENLRMURYCQLUPMSRDLIRIRSRMUHOERETVMBVQKROWAAMBRHRVZGGUWBNGRFVLKHAIJWSXRBY
ATLGHZABNLXHQVLVQMIPWYQWSMVAOATBHPEEJPUHEMBOMMRLOUMBRISIRAZAHFRWLVEBGRXWNUIFSBNLBMFXIJUMHYHOTKXRSMDIHXVE
CBVMNKCZRMIRAHVHEFWLGRITFAOZDRNBCRKYCAGIOGZKTXFLUEIRITKWSIMTHTZAHARWEYBTJYVNUDXGRZYVGQBPUPAMBLKVRDFCRBUYX
ZHVIJHKNHGISIIFIQBRUJKIIRIXLRIHLVEMOZTYHUSGRITTTFLSEYIWHMYQQIOGBUHEUXHEJPIXSXNZDRJVWTHLBELPCIWOJXQNVITU
WOUZUYOXYEWANACGLVRYTYHLOFXECOMZDXADPRXMUMZFVXVVAQTNBPVWIRALGMHLRVXFSDKYUAWKIRMTZMBGEDNVYPHBIYELQKSWCA
JWNTNZRLMCGVLFTJIXUADABJIXKLCTKXSHBYAKUAJICLQZYOYEBWFBBNPGAVWMJFIOKMBNWXZZMYAHJFIPFJQTNPCGKMEGPOZFUAVMFABN
LLWNUPVGLKCBFZDRJVWTPMMARXRZMZHBVUVVZQRPTMPDVCBGGTGCLVVJIYSHIXLXSWOXLCHVJBVGFSUVGLRIMEVNUYUIRQGGJVGYGKE
KNAOHLERGSROEVNLCDBUVBSFEUQBFLGQIVNFQXXRBCOTNNFXLNTFRFRIHPIRABZVNHRSRVKHVGAXFLVLVJIYHIJNUIEGTEBGMHFGVFALBE
UGFSDZCTPVUGLRXJQZOMBRKYDNVHLYIEHLVENOUXCYOHVBSFLYYIWRXCEHLILRYUBLGPESFWYNRTRDAUGHZVEGUBPUAWSSIQUBLFLVLVU
IJUHNHIEBZJLKYQWSJGIEOHGRWLVIKOAUFQSTUWQJXVHWXFNBLZLGXUHVDAZYXMFWDWRCZUAXWGLSEPPGYFSBXOEBEZOTNLRYNBVZIXUO
```

```

OIKBLUHSGSKMETILAXLFXRIVAKZAYFDMUVVNLKBRDVKBVYZHJRDGVSCRHGXRXYVVMGPNUXZPCRHKCFKEGCMTPGAFRWYRTRQNMGJSSNKQ
OHGRRRTBRLNLCFWVVFAMLMGLVVBQBNLFUVGXYBCMOMFVJLKYGVUMVIHUVWGPAMIBZEJYMLANHFSSBRVTVHNUHVGYYIILBHTHRJBSEVVUAPE
KPPZOXMVOIEPMUMMBVVTCKNKKOHRQYIIVBOZGNOOIRXQZOTMNSIRPMLBEUGPSJCPKYXFNIRFTKLICAJWFHTZOXLRLEWBAONGISDRPQQYAN
LODRTRVGANLRFVFFHVZFTHQIWEYOBHTZMEQQYAAFYFRPVFWAYVYBIWFHVJAACFLWKUMTLMBRUAFETJAAAYDRUBNZOXRDHKUMXLBMWECV
DOUBZBPEENVEDAYEHXFOMLVNHQWLVTUPTZMMGKIIRPUDXPRUIEWECIHPNNZLKFVIIJBTGKRSHBYBBATVMUJDVBVNBZOBMGKIDBAZCBPV
GEEQTGYZYFWTCNKPGNULWHHQAPIOHKUMNHDOTBSBHZUBRIHPYBBVVHDONJEIQMTLKQNVTFALKYBHTZLVGPKYMGHPXUMXTBMGUIJFIH
VNNNPSJGZKJXHGRLTVLKMUGWLRGUTXHGVLVSWAUWBRUPRQGNLTXXVQKKBEGYWBURKFBZPFCAJEYYIJFROLXOFLWATNUEHCFAZPEFVJR
FEQTNMBNWIYJPGAWILRYDRITIRNUDXNUIZFHOEHRFGMBLGUIJVVBNNOAMBNWNUGEVNLRUEXTMJHUIHWCFFHZJBEFAHWJFPKKHYFQXIRUKTU
YEEIZAOXHZARGFLGQZZXYVZVYBNLKFNGCANNAGDTLRRJKUMZYHO00IRFEKSEMBBSLJMXLMUYNMETIHVNNGKIDBWTPTGPHVKNQTSRUJDVVB
NZOTNLRYFATEZTCQWLRGITKLIFXHURVRFBNYRSBFTORXHBRRVFBGRBHTDGKQVWZHQUBHFABEVNBNIYRTWEVNGHHLYJPEZHDHVXBVLJPGAL
RYIRBUVNHEHPZNJRLTHLZEPGGAOZDRNRAGDMBRXRYRZJHRQNVFVBKYPYOMNVTRKHCGWLAVUDTSCOIRFMJVGNOHWLPPGTUUALIZFPG
SEABWSFVZKHEFLPIRABCOCTNVEZQIHNHQQVYQGIBFVWCKUWANASHBYBBAGPZSBXNZGPKNAIFWPPZQYAKYFVSWGPKZTCTBSLWQLHFCYBV
RTOKKRIPYNNQZKHSBXLIRIMGURCQHENUMXLMITRQPYIJFHZPRYIFMGSNHBIXYRUGZPYFKSFGAZBYZQRAEJMRSPRQXLNTRFYCAGECRIJAA
UGVAYRZKFOEHAIBVMSTXLBYPHSUFHOEHPGWUTNURRINEVVPYEDRUUIBLUQDMDFWOAMUXHWVIMTSHHTVPCYOHYGGFAKKUMLNWIPIB
CXHMNFODBZKSHIXBSLEJGJDCFVSLALMKXHQHPBCSBROLXOFJIIYXYUOCNBZPXXNESLGGUBFOTRMTWAATFBQIJBANLWYPLHVBUATER
DGKVTIRBRUWVYNYPGWRWLVBXVVGAYWMCWCLKZHOIEBCMOMIPDYJRAAJAUQLWRFBKYTNDRQFSBCVPIHOHEGJKHUUQLHVNQZDTMCHETRNA
SHLVWPFBSKKIYNIWHTNVPIYHVRYTEVNNLAVEMAWLYGZLGRAXLTFYBLRCKXUBHTLWKUIZDBNURYKNVEVGYGDOZAOTVMCPHWLQLKUESGKI
WHTRTHIAKEJQQYHIJRDVVQNXVFARQWFXGUGZGCTKXVIMTVGNUMHRZTPZVYUFTGBAKKYOYOQFBVGIBNBIQZFAOUZVJLKCZCXHGMKSZUTU
YVQKWHTRPMMARAFALKYMBNWLZITZTLRQXRJIXLHZGKIJVBAHMCBQXYBCMOBNFRCLBNLMCALIJGJOAHZQLJWRZKUVYSRVEBNBFAUFLFJ
MBLKUZRSEGPGBAMAWJLYTJVMXWJLAKZPHHNWECYQZZTHRVTPVQGESPEUMKVKGSIIBEPVZNUYMBBVINUWJPLFVNIJHVRPZBGDLLZITHGXL
RYBNQGAUXZJRJKJWBLGNHUIUBCZPGNBJIEFWQFHMZLHEVONAAIHUWKZKAKCRYIKUMSPLMVQKWEIMTXHGRJKUMSVHNNQHIRAZVKYGIWHTR
THIAWLVEOSEMGRTKUMTPZBGLJERMJIXYIHRZSQZIXWBPIJNVOTIYELWYNJRLGCTKXRFAATFYEGVRJAZVTWYRWVAVZTNWUWMDRZTTCVAF
VSWXLMRKEIIMYAFIBQXYRBKHFISKYDNVUGUWSEORVAZVINUHRZTPZHFBZMETITJBYAWZZEOASTNRRFARKZHICYHSWEWAUWMIOMITGAOUM
BRVMXUBGZFUYYOKVZIOBHVKFEJGMKUTLPVFWASTHQWLVYIXNXMGVTYRZKOHQVWVYRLUPGAGKIJRLGFLMUHAFALKYXXNVWYRAZHXYQDXKU
MYLHVWHGKFBOTXXBHWEBSLHQVQXYVAVSTWRKMJGWXFKYCEKFKQZZXFSVLVGWUDTMVQKVAARRI

```

Langkah Penyelesaian

Vigenère Cipher rentan terhadap analisis **Kasiski Examination** jika kunci yang digunakan relatif pendek. Metode ini mengasumsikan bahwa pengulangan rangkaian karakter (*substring*) pada *ciphertext* terjadi ketika *substring plaintext* yang sama dienkripsi menggunakan bagian kunci yang sama.

Penyelesaian soal ini dibagi menjadi dua tahap: **identifikasi panjang kunci** dengan *Kasiski Examination*, lalu **pencarian kunci** dengan *Frequency Analysis* bahasa Inggris.

Tahap 1: *Kasiski Examination*

1. Cari semua rangkaian karakter yang berulang pada *ciphertext* dengan panjang minimal 3 karakter (`VigenereSolver.find_repeated_sequences` pada `src/22/solver.py`).
2. Hitung jarak antarkemunculan dari masing-masing *substring* berulang tersebut.
3. Cari faktor dari setiap jarak tersebut dan hitung frekuensi kemunculan setiap faktor (`VigenereSolver.get_kasiski_factors`). Faktor yang paling sering muncul berpeluang besar merupakan panjang kunci atau kelipatannya.

Hasil pencarian jarak dan pemfaktoran menunjukkan beberapa faktor yang dominan:

Faktor	Frekuensi Muncul
2	2144
5	1898
10	1866
4	1014
20	888

Tiga faktor teratas yang paling sering muncul berturut-turut adalah 2, 5, dan 10. Mengambil nilai maksimum dari sekumpulan faktor dengan frekuensi tertinggi ini memberikan tebakan panjang kunci yang sangat meyakinkan, yaitu **10**.

Tahap 2: Frequency Analysis

Setelah panjang kunci $L = 10$ diketahui, *ciphertext* dapat dibagi menjadi L bagian (*sub-text*), di mana *sub-text* ke- i berisi huruf-huruf pada posisi $i, i + L, i + 2L, \dots$ dari *ciphertext*. Karena setiap *sub-text* dienkripsi dengan satu karakter kunci yang konstan, *cipher* ini dapat dipandang sebagai gabungan L buah sandi **Caesar Cipher**.

1. Untuk setiap *sub-text*, hitung frekuensi kemunculan hurufnya.
2. Lakukan uji coba pergeseran (*shift*) dari 0 hingga 25.
3. Untuk setiap pergeseran, hitung *dot product* antara frekuensi huruf yang dihasilkan dengan probabilitas frekuensi huruf tunggal (*monogram*) bahasa Inggris (MONOGRAM pada `src/frequencies.py`).
4. Pilih pergeseran dengan nilai *dot product* terbesar sebagai huruf ke- i dari kunci (`VigenereSolver.guess_key`).

Berdasarkan analisis tersebut, kunci sepanjang 10 karakter berhasil ditebak, yaitu UNDERNIGHT.

Artefak

Skrip *solver* yang digunakan untuk melakukan *Kasiski Examination*, menemukan kunci dengan analisis frekuensi, mendekripsi *ciphertext*, serta menyimpannya ke berkas log.

```

1 import json
2 import os
3 import sys
4 from datetime import datetime
5 from typing import Dict, List, Tuple
6 from collections import Counter
7
8 sys.path.insert(0, os.path.dirname(os.path.dirname(os.path.abspath(__file__))))
9 from frequencies import MONOGRAM
10
11 class VigenereSolver:
12     def __init__(self):
13         self.english_freq = {c: pct / 100.0 for c, pct in MONOGRAM.items()}
14
15     def find_repeated_sequences(self, text: str, min_len: int = 3, max_len: int =
16 6) -> Dict[str, List[int]]:
17         seq_positions = {}
18         for length in range(min_len, max_len + 1):
19             for i in range(len(text) - length + 1):
20                 seq = text[i:i+length]
21                 if seq not in seq_positions:
22                     seq_positions[seq] = []
23                 seq_positions[seq].append(i)
24
25         repeated_seqs = {seq: pos for seq, pos in seq_positions.items() if len(pos)
26 > 1}
27         return repeated_seqs
28
29     def get_kasiski_factors(self, text: str, max_factor: int = 20) -> List[Tuple[
30 int, int]]:
31         repeated_seqs = self.find_repeated_sequences(text)
32         distances = []
33         for seq, pos in repeated_seqs.items():
34             for i in range(len(pos) - 1):
35                 distances.append(pos[i+1] - pos[i])
36
37         factor_counts = Counter()
38         for dist in distances:

```

```

36         for factor in range(2, max_factor + 1):
37             if dist % factor == 0:
38                 factor_counts[factor] += 1
39
40         return factor_counts.most_common(10)
41
42     def guess_key(self, text: str, key_length: int) -> str:
43         key = ""
44         for i in range(key_length):
45             subtext = text[i::key_length]
46             counts = Counter(subtext)
47             N = len(subtext)
48
49             best_shift = 0
50             max_dot_product = -1
51
52             for shift in range(26):
53                 dot_product = 0
54                 for c in "ABCDEFGHIJKLMNOPQRSTUVWXYZ":
55                     shifted_c = chr((ord(c) - 65 + shift) % 26 + 65)
56                     observed_freq = counts[shifted_c] / N if N > 0 else 0
57                     expected_freq = self.english_freq.get(c, 0.0)
58
59                     dot_product += observed_freq * expected_freq
60
61                 if dot_product > max_dot_product:
62                     max_dot_product = dot_product
63                     best_shift = shift
64
65             key += chr(best_shift + 65)
66         return key
67
68     def decrypt(self, text: str, key: str) -> str:
69         decrypted = []
70         key_len = len(key)
71         key_idx = 0
72         for char in text:
73             if char.isalpha():
74                 shift = ord(key[key_idx % key_len].upper()) - 65
75                 decrypted_char = chr((ord(char.upper()) - 65 - shift) % 26 + 65)
76                 decrypted.append(decrypted_char)
77                 key_idx += 1
78             else:
79                 decrypted.append(char)
80         return "".join(decrypted)
81
82     def load_ciphertext(base_dir: str) -> str:
83         with open(os.path.join(base_dir, "v.txt")) as f:
84             return f.read().strip().replace('\n', '').replace(' ', '')
85
86     def write_plain(base_dir: str, timestamp: str, decrypted: str) -> str:
87         path = os.path.join(base_dir, "log", "plain", timestamp + ".txt")
88         os.makedirs(os.path.dirname(path), exist_ok=True)
89         with open(path, "w") as f:
90             f.write(decrypted)
91         return path
92
93     if __name__ == "__main__":
94         base_dir = os.path.dirname(os.path.abspath(__file__))

```

```

95     content = load_ciphertext(base_dir)
96
97     solver = VigenereSolver()
98
99     print("1. Kasiski Examination")
100     factors = solver.get_kasiski_factors(content)
101     print(f"Faktor yang paling sering muncul dari jarak antar substring berulang:")
102     for factor, count in factors:
103         print(f"Faktor {factor}: {count} kali")
104
105     # Ambil nilai maksimum dari 3 faktor yang paling sering muncul
106     best_key_length = max([factor for factor, count in factors[:3]])
107     print(f"\nPanjang Kunci yang Ditebak (dari Kasiski): {best_key_length}")
108
109     print("\n2. Frequency Analysis")
110     guessed_key = solver.guess_key(content, best_key_length)
111     print(f"Kunci yang Ditebak: {guessed_key}")
112
113     decrypted = solver.decrypt(content, guessed_key)
114
115     print("\n3. Plaintext Preview")
116     print(decrypted[:200] + "...")
117
118     timestamp = datetime.now().strftime("%Y-%m-%d_%H%M%S")
119     plain_path = write_plain(base_dir, timestamp, decrypted)
120
121     print(f"\n[tersimpan ke {os.path.relpath(plain_path)}]")

```

src/22/solver.py

Plaintext Hasil Dekripsi

Kunci

Panjang kunci: 10

Kunci hasil rekonstruksi: UNDERNIGHT

Referensi

Tabel frekuensi *monogram* bahasa Inggris yang digunakan pada Tahap 2 mengacu pada [3]. Referensi mengenai konsep dasar dekripsi Vigenère *Cipher* mengacu pada [6].

2.3 Domain of Hopes and Dreams

Deskripsi Soal

Dalam suatu catatan simposium, Marisa menemukan rekaman mengenai keberadaan sebuah dimensi buatan yang terisolasi dari dunia luar. Batas-batas menuju wilayah rahasia ini dikunci menggunakan segel jimat yang dienkripsi dengan **Playfair Cipher** agar tidak dapat diakses secara sembarangan.

Tugas kami adalah memulihkan naskah petunjuk akses tersebut. Susun kembali matriks kunci dengan menganalisis pola kemunculan pasangan huruf (*bigram* maupun *n-gram* lainnya), distribusi karakter, serta

tebakan kata-kata umum dalam **Bahasa Inggris**. Setelah matriks kunci berhasil direkonstruksi, dekripsikan seluruh dokumen rahasia tersebut.

Berkas lampiran:

- *Attachment:* <https://drive.google.com/file/d/10Dljf85XJXS41PSSqrfaqU0QywTuneuX/view?usp=sharing>
- md5sum: b6f21903cbdb62ca899d6bb2694814b1
- sha256sum: e4bb0e2a2ac8425ef0133b78dd0103308f781d2afdfcc46712b2a23efb7ee597

Berkas *Ciphertext*

```

ONSFLRQLEDQLUKBVUNIBGBUCXMMZTVMSTUKDSVGIPPQMSATNUGCDSBFIAGCYGVSPATGSZDIFNUKBFBELSDYIESQBFTUWTIYMUURTYDSON
SFNTNSIDTUFBLTIAADGNVLTIFBPDMAUBDEMISIVZUNLBTRUBVOIACVTOSSFBFBYQSSFGBCBUCRTDSQINWUHCUYSGPMZTFFBQLFTLPUATA
BANFBESDIAUDSYGSFPZYASIMPOSITLYYPAAQTVIDSZGLLOLPUQNTUIAIACPADYGHCUYSGPSYUBDQUAVGNPTVFIPAMNGIUOLSTOIFNSTOI
YBLQLYUISBGPSTBZIDSWISZTINDILYTLGLSDULCBDILYTLBUUTYGTGEBYMPPOSYULSBMDCYVTTYICBTGBVTOLMETUICVTYFTUKLFUZDU
KPDSCBTBFBGPASZSQBFTIZSUBIAEAPWSLIYFIYCNQIOMSLASZOLNWUMQAGBIZUAVUYADINHNUGCBFTITITAYFIYFTUKPQUOBFUCUUBFIIF
IZBRTFFBGBUAFSAQNEFTPAIFNQUTUDIZLSULOQNAIQPOUOIQNLISIAUQBEGBMFSTUCPAMNGIUPSADIAULYIINUMBFUAQPKISDFLUKELFI
FLICMDIEUISLICTVYEBTEMQDUSRZSSBQMYTSQSDGBVZVTNUFUPAIFAQUOUILHOUGNVTLKNNRSTNRUISBGBBKUTKSSTTYDSRFBGPMYAVT
ICBYMNDYICGBYAFIFNSFSYRFGBRUNWNIVGINBFYAHKLSBFBDFLTGSXUTIFLSIFUNNIAPRUDBUGCRDSIDDIACUNUHKEBLPCUECIEMNDUYT
DPIPATTVPMRUBVIFAQUOUIBSMSYIDSPLSUASP NFYAPOVTTYICUMURTVCUYACKBTMGLOLAUESIAQDOLGBWSTFASANOUZMGCVTDBICUMYS
LSIFUNBWINQIMICVCYELFTGTIBBFDLSUFFBTBCYCUPLATBSIDIACUAQYGHCUYSGPMDOUCREBACUAFEAAQOYSLRIOTBEBWSUABTDSQAYS
YODUFNUKSMQWNTFBGBVTITTBICYSGHCUYSGPSYCVBYDSCBPAQYAEBSWITRFBPEIAQLYGIOMPOSITLYNUGCBFTIMWADBFTITITULIFBFTG
TSIFPABCIIEVBVTINZLMYNYGTSFNQUAPAUQBQLNWUBUKTYBFYAHKTIBFIDYSMWIDBKIAVGRNLWIBGBUCXMMZTVMSTUKDSVGIPPQMSYSVB
QYBFTVDUMFUADSQYQDELZBQTCSSSTYDIACIDDATIBFTVSLFIUADBVBYGSXBEESYLBKQUBFTVBNTYDLABKSQELITUCPRISURINIFFLYS
EVFBLDGVPAZQIDFWCVBUODINHCBBRCVCUBFVTVBUUFASIRYRTEVTPRYSMWIDIZGUINUHSAUKSDINHLDYADIVTICMSAYIVPSTBZIDSSI
NYOSYTEAQBDATBEQLDSGMDUQMNSGEBICGBRTSICRBFYTUMRQLEDQLUKBVUNIBGBUCXMMZTVMSTUKDSVGIPPQMSULIFBFZSXZMWIDGY
TVOUZLICYACRBTVDUUMYAINVTDQOUVTNIEVFBGBUCXMMZTVISTGTIVFADNWAMGIIISUFFBLQSTHKFTIYCRDSBNIZTROUNUCRBIGUINIBFW
CVDUVTBFCUBVDPYBSIBFBGATTBICGBYBSQBFTYMPPOSITLYTIBFIDOUSLHKECBGSYFTUBBUBYCUFIBGVTDDTAISNYDSIVYCNDTGLNVGIPPQ
MSSZSYBFFUAIABQVTUTUBFYAHKESYAVBTIFIZFTUKDSVGEIYTTESPTBPABSIDEWFMZOLUPUGIZYFMDPUBVDPYBSIRLNNBIMUOMTCUBV
DPYBSIRSSPBFYABFTYBSIYCSIYCEURDUUPZLMVBUIBGBAIFGBVTINUCXBSPTVTIQLNWIMDISFYQSSSTBSSGTSIFLSSTYITPASQVGIP
PQMCSBITTBIFGBQIDEBFIDBFTVDUVTMSNGIUMWPDILYTLQWSDSOLCXUATAIFQLCRBFBYUATOSQGLOLIPZBDNOSTTNYAOIACDUITUMMRUO
BTBFBGLYIAYSFRGBRUNWNIVGYOCUTEGBTSQDMYUAYBICINZUFVIAEPQIUNUAFESAPOZSYGBFTUWISDUANUGTIATEGBMSNGIUIZDUNWAMVB
TESDNQDUEZDIADIVGSQBFTUXTNOSTTNYAPITYIFSDGBTSGBIZFIDPICCFRZICXUMTFOISLZIBNBEQLFIPDUTESQLPIEDMANIOYBTUGDIAC
LTBUUTYGBFBDIVVTICDLNICKTUOTIAGVZSIYUMLAIFLDUBLGUFEGPRMRGMQATYVGGIPPQMSQLYAPDQSLAUIFZIDSSIPDUTESDNCPAPFRLA
DEASLXTFQLNWBWLYIAYOSYTPRITUKBFTYLOTYDUIIMDIACUAQUACZICXIADQSLAUIFZIDSSFLFISTHKBFLYIAPICVBTUBDQTVIFYQLU
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BFGUYAQNTFGBGLOLIPZBCTUUKVIUBTCSEGBMSNGIUIFVFCVBEGBMSNGIUGCCYZLEDQMUCNDQMBFTVDSSXGYCUDSBFTVMSNGIURSSPVICP
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 MABFGUFIIZEBTEPDARUCKBEGBPYBUYGIFQLCRBFTVBYIECDUNUCUPRYSYBMWTVDYIEBFTVCVGYCUBFYAHKAPDEMOPTPARTWSBTTFUGFS
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 CRDSIOGINNAMZTIMWDQOLZACRDSQAUAVGIPPMQSRSPAPZMAUBFUAFNBANFBEMAURODUNVTRSCVBUTREVTFMIECBUAQETIEIWTBTAR
 ODFISDZIIIPDBACUAHVSZABUMOLCREBSIDSNUYGIZTGUABTNUGTUALZSZABRLBFAZPMVZSTTYSZIASEFLUKIDINIBQLFIPMATDUUMYASLHK
 BPUZBCCVCUBFUAFNUKMEYBYIAUQBTDS CBDPYBLSPSSFINIAQDVGUBPRMTMYEIEOUAHKTVSLCREBBKAUGATYELEIISZEAIEBUMLSPPSS

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BTDSCBELDIRFTVQLRUUKCBCUDSNEGBTCMPOETITNMNTYGBIZFIAPRLPAUMFSIULXTFQLAII BVITIFTTSPNSTDIACLRATIFBSYUZBEQMYTMS
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BQTIIDERNZIAEOYAQNLRQMQLYGIDDIUABTCUNUKEBLPCUDIIRSZYBLRIACRDSTXDUEBLZSZABMWTVBGBVTDNLCDUSNNWNZSRINYOSSF
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VTNUGCBFSYTUICGBYAFIATBVSZABTFFLCXUFEBTFFELBFDUSTICQLFRZSPMYAIAMVUOITCKGULDZSEVFBTFNYUBFIKIAVGBFBESONTDSBF
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PAYGAUCUBIMYSYIAUDDNAIIMMDDINUBQDQYTAITBCSEGBZLSEQMEVFLFIAPWSINDSDPLACRBFBIYIOBFTVGAQITPEBDQSQBFVTCYDYMN
BEQLFIPAUMQSVXGBLKLYIAPTYAICQYTOYIZLQPPSSTOUIQZSKVQYQDAMDFTPAUABFDYIEITTBZMWITAGBIZTABFIOIODUAI RVUADUFIBF
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RLOBSAUKBFTIQSVSYA
```

Langkah Penyelesaian

Playfair Cipher sangat berkaitan dengan penggunaan bigram (karena aturan enkripsinya berpusat pada pemetaan bigram di sebuah matriks kunci 5×5). Kelemahan *cipher* ini:

- Rentan terhadap analisis frekuensi **bigram**, jika teks *cipher* cukup panjang.
- Bigram dan kebalikannya akan menghasilkan *cipher* yang saling berkaitan: jika bigram plaintext *XY* (baik hubungan searah baris, kolom, maupun *rectangle*) dienkripsi menjadi *cipher PQ*, maka bigram kebalikannya *YX* akan dienkripsi menjadi *QP*, kebalikan dari *cipher* sebelumnya.

Penyelesaian soal ini dibagi menjadi dua tahap. Tahap pertama adalah **analisis frekuensi bigram secara manual** untuk menambatkan sebagian huruf pada matriks kunci. Tahap kedua adalah **pencarian otomatis dengan *Simulated Annealing* (SA)** untuk melengkapi sel-sel yang tersisa, karena melanjutkan penebakan manual sel demi sel terlalu lambat.

Tahap 1: Analisis Frekuensi Bigram

1. Hitung frekuensi bigram **non-overlapping** pada *ciphertext* (`Solver.freq_bigram` pada `src/23/solver.py`). Perhitungan ini dilakukan pada *ciphertext* **sebelum** dilakukan substitusi apa pun, sehingga hasilnya tidak berubah walau hipotesis matriks kunci diubah-ubah. Terdapat **392** bigram unik, dengan sepuluh besar sebagai berikut.

Bigram (non-overlapping)				
Peringkat	Cipher	Freq (%)	Referensi	Freq (%)
1	BF	3.24	TH	3.56
2	UA	1.95	HE	3.07
3	GB	1.87	IN	2.43
4	VT	1.62	ER	2.05
5	TV	1.60	AN	1.99
6	YA	1.45	RE	1.85
7	IA	1.34	ON	1.76
8	DS	1.22	AT	1.49
9	ID	1.22	EN	1.45
10	QL	1.21	ND	1.35

2. Susun hipotesis awal dari empat pencocokan berikut:

Cipher	Freq (%)	Dugaan <i>Plain</i>	Dasar Dugaan
BF	3.24	TH	bigram tersering pada kedua sisi
GB	1.87	HE	peringkat 3 cipher vs bigram Inggris tersering kedua
VT	1.62	RE	pasangan resiprokal dengan TV
TV	1.60	ER	pasangan resiprokal dengan VT

UA belum dicocokkan karena cukup sulit memetakan huruf *plain* yang beririsan. Melanjutkan ke VT/TV, keduanya muncul dengan frekuensi yang mirip, sehingga hipotesis selanjutnya adalah memetakan VT/TV ke **RE/ER**, karena memang keduanya merupakan pasangan bolak-balik yang sama-sama sering muncul dalam Bahasa Inggris (2.05% dan 1.85%). Pencocokan GB → **HE** juga terverifikasi lewat kemunculan BG → **EH**.

3. Turunkan hubungan posisi antarkhuruf dari keempat hipotesis tersebut. Ketiganya ternyata merupakan hubungan *rectangle*, sehingga masing-masing mengunci baris dan kolom relatif antarkhuruf:

- BF → **TH**: B sebaris dengan T, dan F sebaris dengan H.
- GB → **HE**: G sebaris dengan H, dan B sebaris dengan E.
- TV → **ER**: T sebaris dengan E, dan V sebaris dengan R.

Gabungan ketiganya membentuk blok T-E-B pada satu baris, F-G-H pada baris lain, serta R-V pada baris ketiga, dengan kolom yang saling bersesuaian. Penambahan TF → **IT** kemudian menempatkan I. Total **9 dari 25** sel berhasil ditambahkan secara manual, dan inilah isi GRID pada `src/23/solver.py`.

Catatan: hipotesis sempat keliru ketika R dan V diletakkan pada baris yang sama dengan T, E, dan B. Susunan itu memaksa BE (0.88%) terbaca sebagai **VB**, sebuah bigram yang praktis tidak muncul dalam Bahasa Inggris. Kontradiksi inilah yang menjadi dasar penolakan hipotesis tersebut.

Hasil tahap ini adalah hipotesis matriks kunci berikut, dengan sel kosong menandakan posisi yang belum dapat disimpulkan dari analisis bigram:

I				
T	E	B		
F	G	H		
R	V			

Tahap 2: *Simulated Annealing*

Melanjutkan tahap 1 secara manual untuk 16 sel sisanya tidak efisien. Seringkali kesalahan posisi satu huruf baru ditemukan beberapa langkah kemudian. Karena itu, sisa pencarian diserahkan pada *Simulated Annealing*.

1. Hasil dekripsi dinilai dengan **fungsi objektif** berupa **log-probabilitas quadgram** (`Solver.score`), yaitu

$$S = \sum_i \log_{10} P(q_i),$$

dengan q_i adalah quadgram ke- i (jendela 4 huruf yang saling tumpang-tindih) pada hasil dekripsi. Nilai P diambil dari tabel **QUADGRAM** pada `src/frequencies.py` (*quadgram* yang tidak terdaftar diberi nilai dasar 0.001 sebagai penalti). *Quadgram* dipilih karena menilai kelayakan rangkaian huruf yang lebih panjang daripada bigram, sehingga lebih tegas membedakan teks Inggris dari teks acak.

2. Satu langkah dilakukan dengan menukar isi dua sel bebas secara acak (`Solver.swap`). Penukaran antarbaris atau antarkolom penuh hanya diaktifkan bila seluruh sel bebas, karena operasi tersebut akan menggeser sel-sel yang sudah dikunci.
3. Solusi tetangga dengan skor lebih tinggi selalu diterima. Kemudian, solusi yang lebih buruk tetap diterima dengan peluang $\exp(\Delta S/T)$, sehingga pencarian dapat keluar dari optimum lokal. Suhu awal $T = 20$ diturunkan bertahap sebesar 0.5 hingga mencapai 0, dengan 2500 percobaan pada tiap tingkat

suhu (total ± 100.000 evaluasi per *restart*). Proses ini diulang sebanyak 2 *restart*, dan skor terbaik yang akan dipakai.

4. Selama pencarian, penilaian hanya dilakukan pada **2400 karakter pertama** *ciphertext*. Ukuran ini sudah cukup untuk membedakan kunci benar dari kunci salah, namun memangkas waktu secara signifikan. Dekripsi penuh baru dijalankan sekali di akhir memakai kunci terbaik.

Hasil

Pencarian konvergen ke skor -6877.2 dengan waktu eksekusi sekitar **2 menit 30 detik**, dan menghasilkan pemetaan lengkap untuk seluruh **392** bigram unik. Pengujian berulang memberi kunci yang sama secara konsisten.

Perlu dicatat bahwa penguncian sembilan sel hasil tahap 1 sangat menentukan. Ketika seluruh 25 sel dibiarkan bebas, pencarian dengan jumlah *restart* yang sama kerap berhenti pada optimum lokal (misalnya pada skor -7098.3) dengan hasil dekripsi yang tidak terbaca. Dengan kata lain, analisis manual dan pencarian otomatis saling melengkapi: analisis manual memperkecil ruang pencarian dari 25 sel menjadi 16 sel, sementara SA menyelesaikan bagian sisanya.

Artefak

Skrip *solver* yang digunakan untuk menghitung frekuensi bigram *ciphertext*, menjalankan *Simulated Annealing* untuk melengkapi matriks kunci, menurunkan pemetaan bigram *cipher* $\rightarrow$ *plain* berdasarkan aturan Playfair, mendekripsi *ciphertext*, serta mencatat setiap hasil dekripsi ke `log/plain/` dan metadatanya (pemetaan, matriks kunci, frekuensi bigram, dan skor akhir) ke `log/metadata/`.

```

1 import json
2 import math
3 import os
4 import random
5 import sys
6 from datetime import datetime
7 from typing import Dict, List, Tuple
8
9 sys.path.insert(0, os.path.dirname(os.path.dirname(os.path.abspath(__file__))))
10 from frequencies import QUADGRAM
11
12 ALPHABET = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"
13
14 _ = 0
15 GRID = [
16     [_, _, _, _, _],
17     ["I", _, _, _, _],
18     ["T", "E", "B", _, _],
19     ["F", "G", "H", _, _],
20     ["R", "V", _, _, _]
21 ]
22
23 class Solver:
24     def freq_bigram(self, string: str) -> Dict:
25         freq_data = {}
26         total_ngrams = len(string) // 2
27         for i in range(0, len(string), 2):
28             ngram = string[i:i+2]
29             freq_data[ngram] = freq_data.get(ngram, 0) + 1
30         for ngram in freq_data:
31             freq_data[ngram] = (freq_data[ngram] / total_ngrams * 100)
32         return dict(sorted(freq_data.items(), key=lambda item: item[1], reverse=
True))

```

```

33
34 def find_pos(self, grid: List, letter: str) -> Tuple[int, int]:
35     for r, row in enumerate(grid):
36         for c, val in enumerate(row):
37             if val == letter:
38                 return r, c
39     return None
40
41 def decrypt_pair(self, grid: List, a: str, b: str) -> str:
42     ra, ca = self.find_pos(grid, a)
43     rb, cb = self.find_pos(grid, b)
44     if ra == rb:
45         return grid[ra][(ca - 1) % 5] + grid[rb][(cb - 1) % 5]
46     if ca == cb:
47         return grid[(ra - 1) % 5][ca] + grid[(rb - 1) % 5][cb]
48     return grid[ra][cb] + grid[rb][ca]
49
50 def build_mapping(self, grid: List, bigrams: List) -> Dict:
51     return {bigram: self.decrypt_pair(grid, bigram[0], bigram[1]) for bigram in
    bigrams}
52
53 def decrypt(self, string: str, mapping: Dict) -> str:
54     return "".join(mapping[string[i:i+2]] for i in range(0, len(string), 2))
55
56 def score(self, plain: str) -> float:
57     total = 0.0
58     for i in range(len(plain) - 3):
59         total += math.log10(QUADGRAM.get(plain[i:i+4], 0.001))
60     return total
61
62 def random_grid(self, grid: List) -> List:
63     used = [val for row in grid for val in row if val != 0]
64     rest = [letter for letter in ALPHABET if letter not in used]
65     random.shuffle(rest)
66     return [[val if val != 0 else rest.pop() for val in row] for row in grid]
67
68 def free_cells(self, grid: List) -> List:
69     return [(r, c) for r in range(5) for c in range(5) if grid[r][c] == 0]
70
71 def swap(self, grid: List, free: List) -> List:
72     new_grid = [row[:] for row in grid]
73     if len(free) == 25 and random.random() < 0.1:
74         a, b = random.sample(range(5), 2)
75         if random.random() < 0.5:
76             new_grid[a], new_grid[b] = new_grid[b], new_grid[a]
77         else:
78             for row in new_grid:
79                 row[a], row[b] = row[b], row[a]
80     return new_grid
81     (ra, ca), (rb, cb) = random.sample(free, 2)
82     new_grid[ra][ca], new_grid[rb][cb] = new_grid[rb][cb], new_grid[ra][ca]
83     return new_grid
84
85 def anneal(self, string: str, bigrams: List, grid: List, temp: float = 20.0,
86             step: float = 0.5, rounds: int = 2500) -> Tuple[List, float]:
87     free = self.free_cells(grid)
88     current = self.random_grid(grid)
89     current_score = self.score(self.decrypt(string, self.build_mapping(current,
    bigrams)))

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90     best, best_score = current, current_score
91     while temp > 0:
92         for _ in range(rounds):
93             trial = self.swap(current, free)
94             trial_score = self.score(self.decrypt(string, self.build_mapping(
trial, bigrams)))
95             delta = trial_score - current_score
96             if delta > 0 or random.random() < math.exp(delta / temp):
97                 current, current_score = trial, trial_score
98                 if current_score > best_score:
99                     best, best_score = current, current_score
100         temp -= step
101     return best, best_score
102
103     def solve(self, string: str, grid: List, restarts: int = 2) -> Tuple[List,
float]:
104         sample = string[:2400]
105         bigrams = list(self.freq_bigram(sample))
106         best, best_score = None, -math.inf
107         for _ in range(restarts):
108             result, result_score = self.anneal(sample, bigrams, grid)
109             if result_score > best_score:
110                 best, best_score = result, result_score
111         return best, best_score
112
113     def load_ciphertext(base_dir: str) -> str:
114         with open(os.path.join(base_dir, "f.txt")) as f:
115             return f.read().strip()
116
117     def write_plain(base_dir: str, timestamp: str, decrypted: str) -> str:
118         path = os.path.join(base_dir, "log", "plain", timestamp + ".txt")
119         os.makedirs(os.path.dirname(path), exist_ok=True)
120         with open(path, "w") as f:
121             f.write(decrypted)
122         return path
123
124     def write_metadata(base_dir: str, timestamp: str, mapping: Dict, grid_manual: List,
125                       grid_filled: List, freq: Dict, score: float) -> str:
126         path = os.path.join(base_dir, "log", "metadata", timestamp + ".json")
127         os.makedirs(os.path.dirname(path), exist_ok=True)
128         payload = {
129             "timestamp": datetime.now().isoformat(timespec="seconds"),
130             "mapping": mapping,
131             "grid_manual": grid_manual,
132             "grid_filled": grid_filled,
133             "freq_ngram": freq,
134             "score": score,
135         }
136         with open(path, "w") as f:
137             json.dump(payload, f, indent=2)
138         return path
139
140     if __name__ == "__main__":
141         base_dir = os.path.dirname(os.path.abspath(__file__))
142         content = load_ciphertext(base_dir)
143
144         solver = Solver()
145         freq = solver.freq_bigram(content)
146         grid, score = solver.solve(content, GRID)

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```

147 mapping = solver.build_mapping(grid, freq)
148 decrypted = solver.decrypt(content, mapping)
149
150
151 timestamp = datetime.now().strftime("%Y-%m-%d_%H%M%S")
152 plain_path = write_plain(base_dir, timestamp, decrypted)
153 metadata_path = write_metadata(base_dir, timestamp, mapping, GRID, grid, freq,
154                                score)
155 print(decrypted)

```

src/23/solver.py

Plaintext Hasil Dekripsi

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GASIWEREHAVINFUNISEXETHENPLEASEFORGETABOUTWHATISAIDWAITWHATWEREYATALKINABOUTLAUGHTHEREASONKANAKOCAMETOGENS OKYOANYWAYIWASHAVINAHARDTIMEFOLXLLOWINTHATHONESTLYIDONTGIVEAFLIPABOUTTHEHUMANSOFTHEVILXLGETALKABOUTYERSELV ESTHETHEMEWASWHATAREYOUBRINGINGTOGENSOKYOAFTERALLSOFIRSTOFALLWHYDYAMOVETOGENSOKYOANYWAYICAMETOGENSOKYOTOSE EKNEWFAITHINTHEOUTSIDEWORLDSDRINESHAVEBECOMEMERETOUISTXTTRAPSORCALLEDNONSENSICALTHINGSLIKEPOWERSPOTSSOPLAC ESFORHUMANSTOXOFFERTHEIRFAITHTOTHEGODSHAVEDISAPXPEAREDGODSAREBEINGSTHATWILLCEASETOEXISTWITHOUTHUMANSBELIEV INGINTHEMISOYASAYBUTIDONTTHINKYAVEGOTTENMUCHFAITHINGENSOKYOEIFHERIUSTBELIEVINGTHATWEREREALISENOUGHINTHEOUTS IDEWORLDTHEYVESTOPPEDBELIEVINGTHATGODSEVENEXISTINDEXEDEVERYTHINGREALLYCHANGEDONCEWECAMETOGENSOKYOHREHUMAN SBELIEVETHATGODSEXISTASANEVERYDAYFACTWWELXLEVENIFIBELIEVEDINYAORNOTYERRIGH THEREAINTYATHATSNOTGOXODENOUGHFO RTHEOUTSIDEWORLDDEVENIFYOUPERFORMMIRACLERIGHTBEFORETHEIREYESTHEYLXLCONVINCE THEMSELVESITSANILLUSIONANDABSOL UTELYREFUSETOACXCEPTTHATITACTUALLYHAPXPENEDHUMANSAREEASILYSWAYEDCREATURESXSOTHATSTHELATESTFADAFADYESANENTI RELYINTENTIONALFADBUTYOUKNOWTHEBESTPARTISTHATEVENWHILETHEYDONTTHINKTHATGODSORYOUKAEXISTTHEYSTILLBELIEVEIN ALIENSOHNEVERHEARDABOUTTHATTHEMOMENTXTHATHUMANSREALIZEDTHATTHEVASTNESSOFSPACEWASBEYONDTHEIRABILITYTOCONTR OLTHEYSTARTEDTOCONVINCE THEMSELVESTHATX THERE MUST BEALIENSXSOMEWHEREOUTTHEREXEVEN THOUGHNOBODYS CONFIRMEDITYETR ATIONALISMSUDXDENLYTURNSTOMYSTICISMWHENITGOESTOXOFARFUNXNYISNTITIHAVEBEXENCURIQUSABOUTTHISFORAWHILENOWBUTY OURSHRINEHASTWOGODSAMIRIGHTWHYISTHATTECHNICALXLYWEVEGOTXTHREEBUTNEVERMINDTHATFORNOWOURSHRINESBEXENTHROUGHQ UITEABITWELXLIUSTLETMESAYSUWAKOISAGOODPARTNEROOHIKNOWTHISONESUWAKOSTHEONEWHOREALLYRULESTHESHRINERIGHTANDYE RIUSTXTHEONERUNXNINSTUFXFOUTINFRONTAHEMALTHOUGHTHATSNOTQUITEHOWITWORKSITSNOTWRONGTHATXTHEMORIYASHRINEBELON GSTOSUWAKOBUTSUWAKODOESNTHAVEMUCHINTERESTINGATHERINGFAITHYOUSEESOIHAVENOCCHOICEBUTXTOGATHERFAITHFORUSBYMYSE 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ETELYFORGOTTENSOIVECHANGEDFROMAHUMANTOAGODLIKEEXISTENCETHENTHATSCORXRECTWELXLASFARASIKNOWYOUWEREALREADYR EGARDEDASAGODINSHINTOXOHYESIFORGOTTOMENTIONTHISBUTIDONTNORMALXLYLIVEINGENSOKYOHUHLIVEINSPACEICREATEDMYSE LFLIVINGSOMEWHEREWHERE NOBODY CANBOTHER THEMISTHEESSENCEOFWHAT AHERMITISNOWTHATCHAMENTIONITANOTHERHERMITKNOWA LSO LIVESINAHARDTOREACHPLACEHOWDYADOSOMETHINLIKETHATANYWAYITSSIMPLEIFYOUSTUDYENOUGHOF THEHERMETICARTSWOULDYO ULIKETOBECOME MYDISCIPLEUHEMMETHINKABOUTITYOUCOULD BECOMEIMMORTALOXOHOPROSELYTIZINGNOWAREWEDONTXTHINKILXLT AKETHATLYINGDOWNSWITCHNOWANDILLTHROWINEXCELXLENTLUCKONALLYOURFORTUNESFORTWOMONTHSTHENALXLOWMEASWELLYOUAREA YOUKAILIKEHUMANAFTERALLYOUCERTAINLYAREPOPULARLAUGHAXANYWAYIGUESXSTHATWRAPSUPTHESELFINTRODUCTIONSSOWHATI WAN TXTOASKYALLNOWISWHADDYAWANTXTODOINGENSOKYOFROMNOWONITHINKIDLIKETOLIVEACAREFREXELIFEWITHOUTBEINGDISTURBEDIZ

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WIMDOINGTHISFORYOU

Kunci

Matriks kunci 5×5 hasil rekonstruksi:

S	Y	M	P	O
I	U	L	N	A
T	E	B	C	D
F	G	H	K	Q
R	V	W	X	Z

Susunan ini mengikuti pola pembentukan kunci Playfair pada umumnya: deretan kata kunci SYMPOIULNATE di awal, lalu dilanjutkan sisa alfabet secara berurutan (BCDFGHKQVRWXZ).

Referensi

Tabel referensi frekuensi bigram Bahasa Inggris yang digunakan pada Tahap 1 mengacu pada [3, 4]. Tabel QUADGRAM yang dipakai sebagai fungsi objektif pada Tahap 2 disusun dari 45 *quadgram* tersering pada berkas statistik `english_quadgrams.txt` [5], dengan nilai persentase dihitung terhadap total kemunculan seluruh *quadgram* pada berkas tersebut.

2.4 Trinitarian Rhapsody

Deskripsi Soal

Di antara tumpukan lembar catatan risetnya, Marisa menemukan sebuah naskah analisis yang mengulas entitas penguasa tiga alam, sosok yang skalanya melampaui siapa pun di Gensokyo maupun *Lunar Capital*. Naskah yang menyimpan penjelasan mengenai eksistensi mutlak tersebut tidak dapat dibaca karena teksnya telah dienkripsi menggunakan **Affine Cipher**.

Bantulah Marisa mengungkapkan pesan tersembunyi tersebut tanpa bermodalkan kunci awal maupun *known-plaintext*. Manfaatkan sifat matematis aritmatika modulo pada *Affine Cipher* beserta analisis frekuensi

Berkas lampiran:

- Berkas *Ciphertext*

2026/2027

```

OJJVUUQIWNNUDCJWNUECVVWNUCZQHUMWUWJHOKMJUEOHKMGUWKMZILOBUVUWWOLNUZMVBUCLNHUKOLNUHWOZVIQNCVFWNUCWNZOZOHUFMK
OZYWLMVVLNUFUMLNVUWYOFWMZFNUWOZJQHOZOWKMFUNUHMZAHWUOJLNUIOAZYENOMJLUHLNMLFMIWMEELNLNUCHUIUWLNNUVCYNLOJM
VVWUUCZYFMEZWOJHOKLNUBUYCZCZYWNUCWMZAHWUOJLNUIOAZYMFZLNNUWUMHUNUHNZOZAHWNNUWCOFWCZQVAWCOZMZFDMCWUOJNUQMLUC
ZLNULNUOYOZINMWBWUZZLHOABVUWOKUJOHWQNOVMHWCZLNMLNUWUUKWLONOVFNHHCZNCYNHUYMHFENCVULNULWLCKOZIOJOLNUHEHCLUHW
MZFWAHPCCPCZYUPCFUZQWYAYUWLNMLNLCWKMIMPUBUZZLNUUTQUDLCOZOLNUOHCWLNMLNUWCOFWOHCYCMVPCVVMYUNMFMWABWLM
ZLCMVNUQMLUJOVVOECZYMZFLNMLNCWCZQVAWCOZJNUHHCZLNULNUOYOZIEWMEMIOJMFPCZYLNUHDHULWLCYUBIWDHUMFCZYEHOFOJNUHM
KOZYNCWHUMFUHWMZOLNUHLNUOHCWLNMLNUQMLUEMWMKCVIMNOAWUNOVFOFMZFNABVUNOAWUNOVFEHWNCDQOAVFNMPUBUZZKOHUUDH
PMWCPUMZFIULZOLKUZLCOZUFMWKAQNMWLUKDVUEOHWCDCZLNNUZWNQMLUMVOZYECLNXUAWNNUHUKUWMLNUZMNWCLCMZFMDOVVOEUHUPUH
ICKDOHLMZCZFMCVIVCJUMWLNUIEUHULNUKMCZYOFWJLNUOAWUNOVFNUEUPUHCLCWQVUMHLNMLNUWUQCMVDWCLCOZYCPUZLONUQML
UBIXUAWCWADNUVFLNHOAYNOALNUHNCWLOHIBIFUDCQLCOZJWOAZFOZQOCZWOJNUQMLUOZLNUNMZFOJXUAWMWCYNVCYNLUFCCZKOHUHUQUZ
LHUWUMHQNDHUWUZFUFBIFUWLMZFHMZGCCZUCZLNUNOKUHCQNIKZLOFUKULUHNNUQMLUCWQMVUFLUZFUHNUMHLUFMZUDCLNULDUHNMDCZL
UZFUFLQOKDNMWCXUNUHQOZQUHZECLNLNUFCWMDUMHMZQUOJDHUWUDNOZUENUZWNUMWVCWLUFFUKULUHECLNNUHWMHQNJOHUWUDNOZU
JOVVOECZYNUHMBFAQLCOZBINMFUWYAYUWLCZYLNMLFUKULUHNWNOAVFDUMGLOLNUYOFOJLNUWAZNUVCOWWABWUSAUZLVINUQMLUBUQMKU
DUHWUDNOZUWQOKDMZCOZOZNUHIUMHVIROAHZUILOMZFHOKLNUHUMVKWJNMFWUWHPCZYMWMDWIQNOODKDBUQMAWUOJLNCMWWOQCMLCO
ZNUQMLUEMWOZUOJLNUQNCUJYOFFUWUWUJLNUVUAWCZCMZKIWLHUCUWMOZYWCUFUKULUHMZFDUHWUDNOZUMZFLNUHEMWMUKDVUFUF
CQMLUFLONUHZUMHLNUKMCZWMZQLAMHIMLUVUAWCW

```

Langkah Penyelesaian

Tahap awal analisis adalah menentukan frekuensi kemunculan setiap karakter pada *ciphertext*. Karakter yang paling sering muncul diasumsikan berkorespondensi dengan huruf yang paling sering muncul dalam bahasa Inggris, yakni 'E' dan 'T'. Dua karakter dengan frekuensi tertinggi pada teks ini adalah 'U' dan 'M'. Percobaan awal dilakukan dengan memetakan 'U' → 'E' dan 'M' → 'T' untuk mencari sepasang nilai kunci m dan b . Namun, persamaan kongruensi linear yang dihasilkan dari pemetaan ini tidak menghasilkan nilai pengali m yang relatif prima (*coprime*) terhadap modulus 26, sehingga ditolak.

Pendekatan penyelesaian kemudian dialihkan menggunakan metode ***Brute-Force Fallback***.

Dalam sandi *Affine*, parameter m harus relatif prima dengan 26 agar setiap karakter memiliki pemetaan balik yang unik. Hanya ada 12 nilai m yang memenuhi kondisi ini pada $\mathbb{Z}_{26}$ (yaitu 1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25). Di sisi lain, konstanta pergeseran b dapat berupa nilai bulat apa pun dari 0 hingga 25. Dengan demikian, total keseluruhan kombinasi kunci hanyalah $12 \times 26 = 312$ kemungkinan.

Langkah pencarian *brute-force* yang dilakukan adalah:

1. Untuk setiap 312 kemungkinan pasangan (m, b) , dekripsikan *ciphertext* menggunakan *Affine Cipher*.
2. Hitung skor kelayakan dari setiap teks hasil dekripsi menggunakan nilai frekuensi kemunculan *bigram*, *trigram*, dan *quadgram* dalam korpus bahasa Inggris standar (BIGRAM, TRIGRAM, QUADGRAM pada `src/frequencies.py`).
3. Peringkatkan setiap kombinasi kunci berdasarkan total skor yang diperoleh.

Pasangan kunci yang menghasilkan teks dengan kecocokan n -gram tertinggi dan membentuk *plaintext* yang bisa dibaca adalah $m = 15$ dan $b = 12$.

Artefak

Skrip *solver* yang digunakan untuk melakukan dekripsi dengan percobaan heuristik *known-plaintext* berdasarkan frekuensi karakter dan *brute-force fallback*.

```

1 import json
2 import os
3 import sys
4 from datetime import datetime
5 from typing import Dict, List, Tuple
6
7 sys.path.insert(0, os.path.dirname(os.path.dirname(os.path.abspath(__file__))))

```

```
8 from frequencies import BIGRAM, TRIGRAM, QUADGRAM
9
10 class AffineSolver:
11     def calculate_fitness(self, text: str) -> float:
12         score = 0.0
13         for i in range(len(text) - 1):
14             bg = text[i:i+2]
15             if bg in BIGRAM:
16                 score += BIGRAM[bg]
17         for i in range(len(text) - 2):
18             tg = text[i:i+3]
19             if tg in TRIGRAM:
20                 score += TRIGRAM[tg]
21         for i in range(len(text) - 3):
22             qg = text[i:i+4]
23             if qg in QUADGRAM:
24                 score += QUADGRAM[qg]
25         return score
26
27     def count(self, string: str) -> Dict:
28         data = {}
29         for c in string:
30             if c.isalpha():
31                 data[c] = data.get(c, 0) + 1
32         return dict(sorted(data.items(), key=lambda item: item[1], reverse=True))
33
34     def sort(self, string: str) -> List:
35         return list(self.count(string).keys())
36
37     def mod_inverse(self, a: int, m: int) -> int:
38         for x in range(1, m):
39             if (a * x) % m == 1:
40                 return x
41         return -1
42
43     def is_coprime(self, a: int, m: int) -> bool:
44         def gcd(a, b):
45             while b:
46                 a, b = b, a % b
47             return a
48         return gcd(a, m) == 1
49
50     def solve_keys(self, c1_char: str, c2_char: str, p1_char: str = 'E', p2_char:
51 str = 'T') -> Tuple[int, int]:
52         c1 = ord(c1_char) - ord('A')
53         c2 = ord(c2_char) - ord('A')
54         p1 = ord(p1_char) - ord('A')
55         p2 = ord(p2_char) - ord('A')
56
57         diff_c = (c1 - c2) % 26
58         diff_p = (p1 - p2) % 26
59
60         inv_diff_p = self.mod_inverse(diff_p, 26)
61         if inv_diff_p == -1:
62             return -1, -1
63
64         m = (diff_c * inv_diff_p) % 26
65
```

```

66         if not self.is_coprime(m, 26):
67             return -1, -1
68
69         b = (c1 - m * p1) % 26
70         return m, b
71
72     def decrypt(self, string: str, m: int, b: int) -> str:
73         m_inv = self.mod_inverse(m, 26)
74         if m_inv == -1:
75             return ""
76
77         result = []
78         for c in string:
79             if c.isalpha():
80                 y = ord(c) - ord('A')
81                 x = (m_inv * (y - b)) % 26
82                 result.append(chr(x + ord('A')))
83             else:
84                 result.append(c)
85         return "".join(result)
86
87     def brute_force_fallback(self, string: str) -> List[Tuple[int, int, int, str]]:
88         possible_m = [m for m in range(1, 26) if self.is_coprime(m, 26)]
89         results = []
90         for m in possible_m:
91             for b in range(26):
92                 decrypted = self.decrypt(string, m, b)
93                 score = self.calculate_fitness(decrypted)
94                 results.append((score, m, b, decrypted))
95
96         return sorted(results, key=lambda x: x[0], reverse=True)
97
98
99     def load_ciphertext(base_dir: str) -> str:
100         with open(os.path.join(base_dir, "t.txt")) as f:
101             return f.read().strip()
102
103     def write_plain(base_dir: str, timestamp: str, decrypted: str) -> str:
104         path = os.path.join(base_dir, "log", "plain", timestamp + ".txt")
105         os.makedirs(os.path.dirname(path), exist_ok=True)
106         with open(path, "w") as f:
107             f.write(decrypted)
108         return path
109
110     def write_metadata(base_dir: str, timestamp: str, m: int, b: int) -> str:
111         path = os.path.join(base_dir, "log", "metadata", timestamp + ".json")
112         os.makedirs(os.path.dirname(path), exist_ok=True)
113         payload = {
114             "timestamp": datetime.now().isoformat(timespec="seconds"),
115             "key": {
116                 "m": m,
117                 "b": b
118             }
119         }
120         with open(path, "w") as f:
121             json.dump(payload, f, indent=2)
122         return path
123
124 if __name__ == "__main__":

```

```

125 base_dir = os.path.dirname(os.path.abspath(__file__))
126 content = load_ciphertext(base_dir)
127
128 solver = AffineSolver()
129
130 sorted_chars = solver.sort(content)
131 print("Karakter paling sering muncul di ciphertext:", sorted_chars[:5])
132 c1_char = sorted_chars[0]
133 c2_char = sorted_chars[1]
134 print(f"Mencoba memetakan {c1_char} -> E dan {c2_char} -> T")
135
136 m, b = solver.solve_keys(c1_char, c2_char, 'E', 'T')
137
138 decrypted = ""
139 if m != -1:
140     print(f"Ditemukan kunci: m = {m}, b = {b}")
141     decrypted = solver.decrypt(content, m, b)
142     print("Preview hasil dekripsi:")
143     print(decrypted[:100] + "...")
144 else:
145     print("Pemetaan E dan T tidak menghasilkan m yang valid. Mencoba Brute-
146 force Fallback...")
147     results = solver.brute_force_fallback(content)
148     top_result = results[0]
149     m, b = top_result[1], top_result[2]
150     print(f"Hasil terbaik dari brute-force: m = {m}, b = {b}")
151     decrypted = solver.decrypt(content, m, b)
152     print("Preview hasil dekripsi:")
153     print(decrypted[:100] + "...")
154
155 if decrypted:
156     timestamp = datetime.now().strftime("%Y-%m-%d_%H%M%S")
157     plain_path = write_plain(base_dir, timestamp, decrypted)
158     metadata_path = write_metadata(base_dir, timestamp, m, b)
159     print(f"\n[Teks lengkap tersimpan ke {os.path.relpath(plain_path)}]")
160     print(f"[Metadata tersimpan ke {os.path.relpath(metadata_path)}]")

```

src/24/solver.py

Plaintext Hasil Dekripsi

Kunci

Kunci perkalian (*multiplier*) *m*: **15**

Kunci pergeseran (*shift*) *b*: **12**

Referensi

Tabel *n*-gram bahasa Inggris bersumber dari [4] dan [5]. Referensi mengenai konsep dasar dan implementasi dekripsi algoritma *Affine Cipher* mengacu pada [7].

2.5 Silent Goddess

Deskripsi Soal

Di lembar terakhir catatan risetnya, Marisa menemukan salinan artikel investigasi mengenai sosok dewi misterius bersayap tunggal yang terpergok menyusup ke fasilitas di kedalaman Hutan Bambu. Potongan artikel berita tersebut kini tidak dapat dibaca karena telah dienkripsi menggunakan **Hill Cipher**.

Sang jurnalis yang menyusunnya enggan memberikan matriks kunci secara langsung. Ia hanya membocorkan satu petunjuk bahwa naskah investigasi tersebut diawali dengan kata **“EXPLOSIVE SCOOP”**.

Bantulah Marisa merekonstruksi matriks kunci menggunakan petunjuk awal *plaintext* tersebut melalui **Known Plaintext Attack** (KPA). Hitung invers modulo dari matriks kunci yang ditemukan, lalu gunakan hasilnya untuk mendekripsi sisa *ciphertext* hingga pesan lengkap artikel terungkap.

Berkas lampiran:

- Attachment: <https://drive.google.com/file/d/1aS17ZuSGAwfpcus4F6HEP3wPVXYIHE7n/view?usp=sharing>
- md5sum: 84e86bf1dafb89ec420c114abd60498c
- sha256sum: 6113d33c379b4727e120d3146cfc39ab3004842c0252848dc56c621ba7a5d25f

Berkas Ciphertext

```
VNUXGMMXSASGLRPXOUAIOEKONMNANINLBTMTJFACQMNZAUZIRRYXQOECANINLYACCDKPKICRKFAGZXJKBJOPWJCAISEJCVVECKNINL
BTLPiHPRRFJATIIYQXGJULBVWHMBDPSMEEKGDLUVBAMUIWDOTYJKYZKITJKZJBNBYSQVSSXMWPGFQAWMQBNBDPSMEIMYXJMWOLWGBLOTMM
XTNJQYFNJVRLJHZOLLRCOJHJZADHACNEVTMSPRFJUJZMSPCOGSARZWDDNAIUBMWUKEXLEPIQGZTICEOWWROCRNOBJCDIIBDOHLBVLZMOJ
KSNKKYYQXPFLLLRCNDRMXSCXGKGUYQFZQFXFRHTWPUJQWDOTYJKYZOKLPWJTXPMXGMTKCRZSAINMRYUZEVLBVWHMGPVEFWVUBOOKARSI
VYHACPWJSQYGJHSHGQGOUJRCFBFLJJRJATANPQCBKGPVSUENOBNDRCNUAEYSIXIXNFVQMNSFKKYZWDGPVBIFMKTACPPGJYCJXJMSMHVM
QEGKOYPINBDNADHUVREPKZQVCKRNIIZOLQSZTHUDJCDNTBODMGVPZDXPJXRHHKQSFYXWKZIHDKTCVQJUIUVFCVASASKKIVIVJYDGBSLXC
YFYMWDUIZVBXCOBGCHHADGWHLBVWHMUEDACMKRNONEVBZSHYFFHMYGHTKTXPLMZZFTUGSGPVBOTZWDBRLXRH
```

Langkah Penyelesaian

Hill cipher mudah dipecahkan dengan *known-plaintext attack*. Berdasarkan petunjuk yang diberikan, 14 karakter pertama dari *plaintext* adalah **“EXPLOSIVESCOOP”**. Misalkan untuk *Hill cipher* dengan $m = 3$ diketahui:

- $P = (4, 23, 15) \rightarrow C = (21, 13, 20)$ (dari blok “EXP” $\rightarrow$ “VNU”)
- $P = (11, 14, 18) \rightarrow C = (4, 23, 6)$ (dari blok “LOS” $\rightarrow$ “EXG”)
- $P = (8, 21, 4) \rightarrow C = (12, 12, 23)$ (dari blok “IVE” $\rightarrow$ “MMX”)
- Jadi, $K(4, 23, 15) = (21, 13, 20)$, $K(11, 14, 18) = (4, 23, 6)$, dan $K(8, 21, 4) = (12, 12, 23)$

$$\begin{pmatrix} 21 & 4 & 12 \\ 13 & 23 & 12 \\ 20 & 6 & 23 \end{pmatrix} = K \begin{pmatrix} 4 & 11 & 8 \\ 23 & 14 & 21 \\ 15 & 18 & 4 \end{pmatrix} \pmod{26} \rightarrow K = CP^{-1} \pmod{26}$$

- Matriks balikan dari P adalah $P^{-1} = \begin{pmatrix} 4 & 11 & 8 \\ 23 & 14 & 21 \\ 15 & 18 & 4 \end{pmatrix}^{-1} \pmod{26} = \begin{pmatrix} 8 & 24 & 1 \\ 1 & 0 & 24 \\ 24 & 1 & 25 \end{pmatrix}$
- Sehingga

$$K = CP^{-1} \pmod{26} = \begin{pmatrix} 21 & 4 & 12 \\ 13 & 23 & 12 \\ 20 & 6 & 23 \end{pmatrix} \begin{pmatrix} 8 & 24 & 1 \\ 1 & 0 & 24 \\ 24 & 1 & 25 \end{pmatrix} \pmod{26} = \begin{pmatrix} 18 & 22 & 1 \\ 25 & 12 & 7 \\ 16 & 9 & 11 \end{pmatrix}$$

Selanjutnya, sisa dari pesan *ciphertext* didekripsi secara langsung menggunakan matriks balikan (invers) dari K .

Langkah-langkah Dekripsi

Berikut adalah rincian langkah dekripsi untuk tiap blok karakter (Total 240 blok) menjadi vektor numerik, dikali dengan invers matriks kunci modulo 26, lalu dikembalikan ke bentuk teks:

Blok	Ciphertext	Vektor	Plaintext	Vektor
1	VNU	$(21, 13, 20)^T$	EXP	$(4, 23, 15)^T$
2	EXG	$(4, 23, 6)^T$	LOS	$(11, 14, 18)^T$
3	MMX	$(12, 12, 23)^T$	IVE	$(8, 21, 4)^T$
4	SAS	$(18, 0, 18)^T$	SCO	$(18, 2, 14)^T$
5	CGL	$(2, 6, 11)^T$	OPS	$(14, 15, 18)^T$
6	RPX	$(17, 15, 23)^T$	USP	$(20, 18, 15)^T$
7	OUA	$(14, 20, 0)^T$	ICI	$(8, 2, 8)^T$
8	IOE	$(8, 14, 4)^T$	OUS	$(14, 20, 18)^T$
9	KON	$(10, 14, 13)^T$	SHA	$(18, 7, 0)^T$
10	MNA	$(12, 13, 0)^T$	DYC	$(3, 24, 2)^T$
11	NIN	$(13, 8, 13)^T$	HAR	$(7, 0, 17)^T$
12	LBT	$(11, 1, 19)^T$	ACT	$(0, 2, 19)^T$
13	MWT	$(12, 22, 19)^T$	ERI	$(4, 17, 8)^T$
14	JFA	$(9, 5, 0)^T$	SFR	$(18, 5, 17)^T$
15	CQM	$(2, 16, 12)^T$	EQU	$(4, 16, 20)^T$
16	NZA	$(13, 25, 0)^T$	ENT	$(4, 13, 19)^T$
17	UIZ	$(20, 8, 25)^T$	ING	$(8, 13, 6)^T$
18	IRR	$(8, 17, 17)^T$	PHA	$(15, 7, 0)^T$
19	YXQ	$(24, 23, 16)^T$	RMA	$(17, 12, 0)^T$
20	OEC	$(14, 4, 2)^T$	CEU	$(2, 4, 20)^T$
21	ANI	$(0, 13, 8)^T$	TIC	$(19, 8, 2)^T$
22	NLY	$(13, 11, 24)^T$	ALF	$(0, 11, 5)^T$
23	ACC	$(0, 2, 2)^T$	ACI	$(0, 2, 8)^T$
24	DKP	$(3, 10, 15)^T$	LIT	$(11, 8, 19)^T$
25	KIC	$(10, 8, 2)^T$	YWE	$(24, 22, 4)^T$
26	RKF	$(17, 10, 5)^T$	HAV	$(7, 0, 21)^T$
27	GAZ	$(6, 0, 25)^T$	ELE	$(4, 11, 4)^T$
28	XJK	$(23, 9, 10)^T$	ARN	$(0, 17, 13)^T$
29	BJO	$(1, 9, 14)^T$	EDT	$(4, 3, 19)^T$
30	PWJ	$(15, 22, 9)^T$	HAT	$(7, 0, 19)^T$
31	CAI	$(2, 0, 8)^T$	AMY	$(0, 12, 24)^T$
32	SEJ	$(18, 4, 9)^T$	STE	$(18, 19, 4)^T$
33	CVE	$(2, 21, 4)^T$	RIO	$(17, 8, 14)^T$
34	ECK	$(4, 2, 10)^T$	USC	$(20, 18, 2)^T$
35	NIN	$(13, 8, 13)^T$	HAR	$(7, 0, 17)^T$
36	LBT	$(11, 1, 19)^T$	ACT	$(0, 2, 19)^T$
37	LPI	$(11, 15, 8)^T$	ERH	$(4, 17, 7)^T$
38	HPR	$(7, 15, 17)^T$	ASB	$(0, 18, 1)^T$
39	RFJ	$(17, 5, 9)^T$	EEN	$(4, 4, 13)^T$
40	ATI	$(0, 19, 8)^T$	VIS	$(21, 8, 18)^T$
41	YQX	$(24, 16, 23)^T$	ITI	$(8, 19, 8)^T$
42	GJU	$(6, 9, 20)^T$	NGE	$(13, 6, 4)^T$
lanjut di halaman berikutnya,				

Tabel 2.1 – lanjutan dari halaman sebelumnya

Blok	Ciphertext	Vektor	Plaintext	Vektor
43	LBV	$(11, 1, 21)^T$	IEN	$(8, 4, 13)^T$
44	WHM	$(22, 7, 12)^T$	TEI	$(19, 4, 8)^T$
45	BDP	$(1, 3, 15)^T$	GEN	$(6, 4, 13)^T$
46	SME	$(18, 12, 4)^T$	SOK	$(18, 14, 10)^T$
47	EKG	$(4, 10, 6)^T$	YOS	$(24, 14, 18)^T$
48	DLU	$(3, 11, 20)^T$	ONL	$(14, 13, 11)^T$
49	VBA	$(21, 1, 0)^T$	YDR	$(24, 3, 17)^T$
50	MUI	$(12, 20, 8)^T$	UGO	$(20, 6, 14)^T$
51	WDO	$(22, 3, 14)^T$	RGA	$(17, 6, 0)^T$
52	TYJ	$(19, 24, 9)^T$	NIZ	$(13, 8, 25)^T$
53	KYZ	$(10, 24, 25)^T$	ATI	$(0, 19, 8)^T$
54	KIT	$(10, 8, 19)^T$	ONS	$(14, 13, 18)^T$
55	JKZ	$(9, 10, 25)^T$	HED	$(7, 4, 3)^T$
56	JBN	$(9, 1, 13)^T$	ISP	$(8, 18, 15)^T$
57	OBY	$(14, 1, 24)^T$	LAY	$(11, 0, 24)^T$
58	SQV	$(18, 16, 21)^T$	SFA	$(18, 5, 0)^T$
59	SSX	$(18, 18, 23)^T$	SHI	$(18, 7, 8)^T$
60	MWP	$(12, 22, 15)^T$	ONU	$(14, 13, 20)^T$
61	GFQ	$(6, 5, 16)^T$	NCO	$(13, 2, 14)^T$
62	AWM	$(0, 22, 12)^T$	MMO	$(12, 12, 14)^T$
63	QBN	$(16, 1, 13)^T$	NTO	$(13, 19, 14)^T$
64	BDP	$(1, 3, 15)^T$	GEN	$(6, 4, 13)^T$
65	SME	$(18, 12, 4)^T$	SOK	$(18, 14, 10)^T$
66	IMY	$(8, 12, 24)^T$	YOW	$(24, 14, 22)^T$
67	XJM	$(23, 9, 12)^T$	ITH	$(8, 19, 7)^T$
68	WOL	$(22, 14, 11)^T$	ADI	$(0, 3, 8)^T$
69	WGB	$(22, 6, 1)^T$	STI	$(18, 19, 8)^T$
70	LOT	$(11, 14, 19)^T$	NCT	$(13, 2, 19)^T$
71	MMX	$(12, 12, 23)^T$	IVE	$(8, 21, 4)^T$
72	TNJ	$(19, 13, 9)^T$	SIN	$(18, 8, 13)^T$
73	QYF	$(16, 24, 5)^T$	GLE	$(6, 11, 4)^T$
74	NJV	$(13, 9, 21)^T$	WIN	$(22, 8, 13)^T$
75	RLJ	$(17, 11, 9)^T$	GED	$(6, 4, 3)^T$
76	HZO	$(7, 25, 14)^T$	APP	$(0, 15, 15)^T$
77	LLR	$(11, 11, 17)^T$	EAR	$(4, 0, 17)^T$
78	COJ	$(2, 14, 9)^T$	ANC	$(0, 13, 2)^T$
79	HJZ	$(7, 9, 25)^T$	EAN	$(4, 0, 13)^T$
80	ADH	$(0, 3, 7)^T$	DHA	$(3, 7, 0)^T$
81	ACN	$(0, 2, 13)^T$	SNO	$(18, 13, 14)^T$
82	EVT	$(4, 21, 19)^T$	TBE	$(19, 1, 4)^T$
83	MSP	$(12, 18, 15)^T$	ENS	$(4, 13, 18)^T$
84	RFJ	$(17, 5, 9)^T$	EEN	$(4, 4, 13)^T$
85	UIZ	$(20, 8, 25)^T$	ING	$(8, 13, 6)^T$
86	MSP	$(12, 18, 15)^T$	ENS	$(4, 13, 18)^T$
87	COG	$(2, 14, 6)^T$	OKY	$(14, 10, 24)^T$
88	SAR	$(18, 0, 17)^T$	OBE	$(14, 1, 4)^T$
89	ZWD	$(25, 22, 3)^T$	FOR	$(5, 14, 17)^T$
90	DNA	$(3, 13, 0)^T$	ETH	$(4, 19, 7)^T$
91	IUB	$(8, 20, 1)^T$	ERE	$(4, 17, 4)^T$
lanjut di halaman berikutnya,				

Tabel 2.1 – lanjutan dari halaman sebelumnya

Blok	Ciphertext	Vektor	Plaintext	Vektor
92	MWU	$(12, 22, 20)^T$	ISS	$(8, 18, 18)^T$
93	KEX	$(10, 4, 23)^T$	URE	$(20, 17, 4)^T$
94	LEP	$(11, 4, 15)^T$	LYN	$(11, 24, 13)^T$
95	IQG	$(8, 16, 6)^T$	OWA	$(14, 22, 0)^T$
96	ZTI	$(25, 19, 8)^T$	YTH	$(24, 19, 7)^T$
97	CEO	$(2, 4, 14)^T$	ISI	$(8, 18, 8)^T$
98	WWR	$(22, 22, 17)^T$	SJU	$(18, 9, 20)^T$
99	OCR	$(14, 2, 17)^T$	STA	$(18, 19, 0)^T$
100	NOB	$(13, 14, 1)^T$	NOR	$(13, 14, 17)^T$
101	JCD	$(9, 2, 3)^T$	DIN	$(3, 8, 13)^T$
102	IIB	$(8, 8, 1)^T$	ARY	$(0, 17, 24)^T$
103	DOH	$(3, 14, 7)^T$	PAT	$(15, 0, 19)^T$
104	LBV	$(11, 1, 21)^T$	IEN	$(8, 4, 13)^T$
105	LMZ	$(11, 12, 25)^T$	TIN	$(19, 8, 13)^T$
106	MOJ	$(12, 14, 9)^T$	WHI	$(22, 7, 8)^T$
107	KSN	$(10, 18, 13)^T$	CHC	$(2, 7, 2)^T$
108	KKY	$(10, 10, 24)^T$	ASE	$(0, 18, 4)^T$
109	YQX	$(24, 16, 23)^T$	ITI	$(8, 19, 8)^T$
110	PFL	$(15, 5, 11)^T$	SCL	$(18, 2, 11)^T$
111	LLR	$(11, 11, 17)^T$	EAR	$(4, 0, 17)^T$
112	CND	$(2, 13, 3)^T$	THA	$(19, 7, 0)^T$
113	RMX	$(17, 12, 23)^T$	TSH	$(19, 18, 7)^T$
114	SCX	$(18, 2, 23)^T$	EHA	$(4, 7, 0)^T$
115	GKG	$(6, 10, 6)^T$	SSO	$(18, 18, 14)^T$
116	UYQ	$(20, 24, 16)^T$	MEC	$(12, 4, 2)^T$
117	FZQ	$(5, 25, 16)^T$	ONN	$(14, 13, 13)^T$
118	FXF	$(5, 23, 5)^T$	ECT	$(4, 2, 19)^T$
119	XRH	$(23, 17, 7)^T$	ION	$(8, 14, 13)^T$
120	TWP	$(19, 22, 15)^T$	TOT	$(19, 14, 19)^T$
121	UJQ	$(20, 9, 16)^T$	HEO	$(7, 4, 14)^T$
122	WDO	$(22, 3, 14)^T$	RGA	$(17, 6, 0)^T$
123	TYJ	$(19, 24, 9)^T$	NIZ	$(13, 8, 25)^T$
124	KYZ	$(10, 24, 25)^T$	ATI	$(0, 19, 8)^T$
125	OKL	$(14, 10, 11)^T$	ONW	$(14, 13, 22)^T$
126	PWJ	$(15, 22, 9)^T$	HAT	$(7, 0, 19)^T$
127	TXP	$(19, 23, 15)^T$	CON	$(2, 14, 13)^T$
128	MXG	$(12, 23, 6)^T$	NEC	$(13, 4, 2)^T$
129	MTK	$(12, 19, 10)^T$	TIO	$(19, 8, 14)^T$
130	CRZ	$(2, 17, 25)^T$	NDO	$(13, 3, 14)^T$
131	SAI	$(18, 0, 8)^T$	ESS	$(4, 18, 18)^T$
132	NMR	$(13, 12, 17)^T$	HEH	$(7, 4, 7)^T$
133	YUZ	$(24, 20, 25)^T$	AVE	$(0, 21, 4)^T$
134	EVG	$(4, 21, 6)^T$	TOE	$(19, 14, 4)^T$
135	LBV	$(11, 1, 21)^T$	IEN	$(8, 4, 13)^T$
136	WHM	$(22, 7, 12)^T$	TEI	$(19, 4, 8)^T$
137	GPV	$(6, 15, 21)^T$	THE	$(19, 7, 4)^T$
138	EFW	$(4, 5, 22)^T$	REA	$(17, 4, 0)^T$
139	VUB	$(21, 20, 1)^T$	RER	$(17, 4, 17)^T$
140	OOK	$(14, 14, 10)^T$	UMO	$(20, 12, 14)^T$
lanjut di halaman berikutnya,				

Tabel 2.1 – lanjutan dari halaman sebelumnya

Blok	Ciphertext	Vektor	Plaintext	Vektor
141	ARS	$(0, 17, 18)^T$	RSA	$(17, 18, 0)^T$
142	IVY	$(8, 21, 24)^T$	BOU	$(1, 14, 20)^T$
143	HAC	$(7, 0, 2)^T$	NDT	$(13, 3, 19)^T$
144	PWJ	$(15, 22, 9)^T$	HAT	$(7, 0, 19)^T$
145	SQY	$(18, 16, 24)^T$	EIE	$(4, 8, 4)^T$
146	GJH	$(6, 9, 7)^T$	NTE	$(13, 19, 4)^T$
147	ASH	$(0, 18, 7)^T$	IHO	$(8, 7, 14)^T$
148	GQG	$(6, 16, 6)^T$	USE	$(20, 18, 4)^T$
149	OUJ	$(14, 20, 9)^T$	SLU	$(18, 11, 20)^T$
150	RCF	$(17, 2, 5)^T$	NAR	$(13, 0, 17)^T$
151	BFL	$(1, 5, 11)^T$	IAN	$(8, 0, 13)^T$
152	JJR	$(9, 9, 17)^T$	SWH	$(18, 22, 7)^T$
153	JTA	$(9, 19, 0)^T$	OFL	$(14, 5, 11)^T$
154	NPQ	$(13, 15, 16)^T$	EDF	$(4, 3, 5)^T$
155	CBK	$(2, 1, 10)^T$	ROM	$(17, 14, 12)^T$
156	GPV	$(6, 15, 21)^T$	THE	$(19, 7, 4)^T$
157	SUE	$(18, 20, 4)^T$	MOO	$(12, 14, 14)^T$
158	NOB	$(13, 14, 1)^T$	NOR	$(13, 14, 17)^T$
159	CND	$(2, 13, 3)^T$	THA	$(19, 7, 0)^T$
160	RCN	$(17, 2, 13)^T$	TIT	$(19, 8, 19)^T$
161	UAE	$(20, 0, 4)^T$	ISA	$(8, 18, 0)^T$
162	YSI	$(24, 18, 8)^T$	SEC	$(18, 4, 2)^T$
163	XIX	$(23, 8, 23)^T$	RET	$(17, 4, 19)^T$
164	NFV	$(13, 5, 21)^T$	MIL	$(12, 8, 11)^T$
165	QMN	$(16, 12, 13)^T$	ITA	$(8, 19, 0)^T$
166	DSF	$(3, 18, 5)^T$	RYB	$(17, 24, 1)^T$
167	KKY	$(10, 10, 24)^T$	ASE	$(0, 18, 4)^T$
168	ZWD	$(25, 22, 3)^T$	FOR	$(5, 14, 17)^T$
169	GPV	$(6, 15, 21)^T$	THE	$(19, 7, 4)^T$
170	BIF	$(1, 8, 5)^T$	LUN	$(11, 20, 13)^T$
171	MKT	$(12, 10, 19)^T$	ARC	$(0, 17, 2)^T$
172	ACP	$(0, 2, 15)^T$	API	$(0, 15, 8)^T$
173	PGJ	$(15, 6, 9)^T$	TAL	$(19, 0, 11)^T$
174	YCJ	$(24, 2, 9)^T$	IFE	$(8, 5, 4)^T$
175	XJM	$(23, 9, 12)^T$	ITH	$(8, 19, 7)^T$
176	SMH	$(18, 12, 7)^T$	ERO	$(4, 17, 14)^T$
177	VMQ	$(21, 12, 16)^T$	FTH	$(5, 19, 7)^T$
178	EGK	$(4, 6, 10)^T$	ESE	$(4, 18, 4)^T$
179	OYP	$(14, 24, 15)^T$	ARE	$(0, 17, 4)^T$
180	INB	$(8, 13, 1)^T$	TRU	$(19, 17, 20)^T$
181	DNA	$(3, 13, 0)^T$	ETH	$(4, 19, 7)^T$
182	DHU	$(3, 7, 20)^T$	ENJ	$(4, 13, 9)^T$
183	VRP	$(21, 17, 15)^T$	UST	$(20, 18, 19)^T$
184	EKZ	$(4, 10, 25)^T$	WHA	$(22, 7, 0)^T$
185	QVC	$(16, 21, 2)^T$	TIS	$(19, 8, 18)^T$
186	KRN	$(10, 17, 13)^T$	THI	$(19, 7, 8)^T$
187	IIZ	$(8, 8, 25)^T$	SPE	$(18, 15, 4)^T$
188	OLQ	$(14, 11, 16)^T$	RSO	$(17, 18, 14)^T$
189	SZT	$(18, 25, 19)^T$	NDE	$(13, 3, 4)^T$
lanjut di halaman berikutnya,				

Tabel 2.1 – lanjutan dari halaman sebelumnya

Blok	Ciphertext	Vektor	Plaintext	Vektor
190	HUD	$(7, 20, 3)^T$	PEN	$(15, 4, 13)^T$
191	JCD	$(9, 2, 3)^T$	DIN	$(3, 8, 13)^T$
192	NTB	$(13, 19, 1)^T$	GON	$(6, 14, 13)^T$
193	ODM	$(14, 3, 12)^T$	HOW	$(7, 14, 22)^T$
194	GPV	$(6, 15, 21)^T$	THE	$(19, 7, 4)^T$
195	ZDX	$(25, 3, 23)^T$	SIT	$(18, 8, 19)^T$
196	PJJ	$(15, 9, 9)^T$	UAT	$(20, 0, 19)^T$
197	XRH	$(23, 17, 7)^T$	ION	$(8, 14, 13)^T$
198	HKD	$(7, 10, 3)^T$	DEV	$(3, 4, 21)^T$
199	QSF	$(16, 18, 5)^T$	ELO	$(4, 11, 14)^T$
200	YXW	$(24, 23, 22)^T$	PSI	$(15, 18, 8)^T$
201	KZI	$(10, 25, 8)^T$	TCO	$(19, 2, 14)^T$
202	HDK	$(7, 3, 10)^T$	ULD	$(20, 11, 3)^T$
203	TCV	$(19, 2, 21)^T$	TUR	$(19, 20, 17)^T$
204	QJI	$(16, 9, 8)^T$	NOU	$(13, 14, 20)^T$
205	UVF	$(20, 21, 5)^T$	TTO	$(19, 19, 14)^T$
206	CVA	$(2, 21, 0)^T$	BEA	$(1, 4, 0)^T$
207	SAS	$(18, 0, 18)^T$	SCO	$(18, 2, 14)^T$
208	KKV	$(10, 10, 21)^T$	OPA	$(14, 15, 0)^T$
209	IVY	$(8, 21, 24)^T$	BOU	$(1, 14, 20)^T$
210	JYD	$(9, 24, 3)^T$	TIL	$(19, 8, 11)^T$
211	GBS	$(6, 1, 18)^T$	LEG	$(11, 4, 6)^T$
212	LXC	$(11, 23, 2)^T$	ALD	$(0, 11, 3)^T$
213	YFY	$(24, 5, 24)^T$	RUG	$(17, 20, 6)^T$
214	MWD	$(12, 22, 3)^T$	SBE	$(18, 1, 4)^T$
215	UIZ	$(20, 8, 25)^T$	ING	$(8, 13, 6)^T$
216	VBX	$(21, 1, 23)^T$	MAN	$(12, 0, 13)^T$
217	COB	$(2, 14, 1)^T$	UFA	$(20, 5, 0)^T$
218	GCH	$(6, 2, 7)^T$	CTU	$(2, 19, 20)^T$
219	HAD	$(7, 0, 3)^T$	RED	$(17, 4, 3)^T$
220	GWH	$(6, 22, 7)^T$	ATE	$(0, 19, 4)^T$
221	LBV	$(11, 1, 21)^T$	IEN	$(8, 4, 13)^T$
222	WHM	$(22, 7, 12)^T$	TEI	$(19, 4, 8)^T$
223	UED	$(20, 4, 3)^T$	ORS	$(14, 17, 18)^T$
224	ACM	$(0, 2, 12)^T$	OME	$(14, 12, 4)^T$
225	KRN	$(10, 17, 13)^T$	THI	$(19, 7, 8)^T$
226	ONE	$(14, 13, 4)^T$	NGM	$(13, 6, 12)^T$
227	VBZ	$(21, 1, 25)^T$	UCH	$(20, 2, 7)^T$
228	SHY	$(18, 7, 24)^T$	BIG	$(1, 8, 6)^T$
229	FFH	$(5, 5, 7)^T$	GER	$(6, 4, 17)^T$
230	MYG	$(12, 24, 6)^T$	WEW	$(22, 4, 22)^T$
231	HTK	$(7, 19, 10)^T$	ILL	$(8, 11, 11)^T$
232	TXP	$(19, 23, 15)^T$	CON	$(2, 14, 13)^T$
233	LMZ	$(11, 12, 25)^T$	TIN	$(19, 8, 13)^T$
234	ZFT	$(25, 5, 19)^T$	UET	$(20, 4, 19)^T$
235	UGS	$(20, 6, 18)^T$	OGA	$(14, 6, 0)^T$
236	GPV	$(6, 15, 21)^T$	THE	$(19, 7, 4)^T$
237	BOT	$(1, 14, 19)^T$	RIN	$(17, 8, 13)^T$
238	ZWD	$(25, 22, 3)^T$	FOR	$(5, 14, 17)^T$
lanjut di halaman berikutnya,				

Tabel 2.1 – lanjutan dari halaman sebelumnya

Blok	Ciphertext	Vektor	Plaintext	Vektor
239	BRL	$(1, 17, 11)^T$	MAT	$(12, 0, 19)^T$
240	XRH	$(23, 17, 7)^T$	ION	$(8, 14, 13)^T$

Artefak

Skrip *solver* yang mengimplementasikan pencarian invers matriks secara modular dengan *Known Plaintext Attack* (KPA) pada bahasa pemrograman Python (`src/25/solver.py`). Skrip ini secara otomatis memecah teks menjadi blok $m = 3$ dan melakukan proses matriks balikan (*decryption*). Log penyelesaian (metadata kunci dan *plaintext*) disimpan di `src/25/log`.

Plaintext Hasil Dekripsi

```
EXPLOSIVESCOOPSUSPICIOUSSHADYCHARACTERISFREQUENTINGPHARMACEUTICALFACILITYWEHAVELEARNEDTHATAMYSTERIOUSCHARACTERHASBEENVISITINGEIENTEIGENSOKYOSONLYDRUGORGANIZATIONSHEDISPLAYSFASHIONUNCOMMONTOGENSOKYOWITHADISTINCTIVESINGLEWINGEDAPPEARANCEANDHASNOTBEENSEENINGENSOKYOBFORETHEREISSURELYNOWAYTHISISJUSTANORDINARYPATIENTINWHICHCASEITISCLEARTHATSHEHAS SOME CONNECTION TO THE ORGANIZATIONWHAT CONNECTION DOES SHE HAVE TO EIENTEITHEREARERUMORSABOUTTHATEIENTEIHOUSES LUNARIANS WHO FLED FROM THE MOON NORTHATITISASECRET MILITARY BASE FOR THE LUNAR CAPITALIFEITHER OF THESE ARE TRUE THEN JUSTWHATISTHISPERSONDEPENDINGONHOWTHESITUATIONDEVELOPSITCOULDTURNOUTTOBEASCOOPABOUTILLEGALDRUGSBEINGMANUFACTUREDATEIENTEIOR SOMETHING MUCH BIGGERWE WILL CONTINUE TO GATHER INFORMATION
```

Kunci

Matriks Kunci (K):

$$\begin{pmatrix} 18 & 22 & 1 \\ 25 & 12 & 7 \\ 16 & 9 & 11 \end{pmatrix}$$

Referensi

- Slide Kuliah IF4020 Kriptografi: Kriptografi Klasik Bagian 2 (Materi *Hill Cipher* dan *Known-Plaintext Attack*).
- GeeksforGeeks, *Hill Cipher*. <https://www.geeksforgeeks.org/dsa/hill-cipher/>

Bab 3

Problem Set (Bonus)

3.1 Ex Machina

Deskripsi Soal

Di tengah *hollow night*, sebuah transmisi aneh muncul dari salah satu wilayah yang telah lama ditinggalkan. Sinyalnya berulang secara teratur, tetapi tidak seorang pun mengetahui siapa yang mengirimkannya ataupun sudah berapa lama mesin tersebut aktif.

Setelah diteliti, pesan tersebut diketahui telah dienkripsi menggunakan **Enigma Machine M3**. Namun, setiap percobaan untuk mendekripsinya menggunakan konfigurasi Enigma M3 selalu menghasilkan teks yang tidak masuk akal. Entah karena pengaruh *hollow night*, kerusakan pada mekanismenya, atau memang karena mesin tersebut sejak awal dibangun dengan aturan yang berbeda, Enigma ini telah dimodifikasi.

Analisislah perilaku mesin tersebut, **temukan perubahan yang membedakannya dari Enigma M3 standar**, kemudian lakukan **kriptanalisis** untuk merekonstruksi konfigurasi yang digunakan dan memulihkan pesan aslinya sebelum *hollow night* berakhir.

Berkas lampiran:

- Attachment: https://drive.google.com/file/d/1TcBU6s876b5ah9EVsuX_xHDTc_0LaWsI/view?usp=sharing
- md5sum: 1074d1cfcaf2482f7229c8f7d40ba097
- sha256sum: fd31f4b0b6475fc15926caea6b28179ea2a1228fe9f063cad8b2d5c215d9031b
- Kode mesin: src/31/enigma-machine/

Berkas *Ciphertext*

```
UTJX CENI RYOW WSVU YEHC YRPR BJWZ NXDJ UNVX WXFE
WOUO DQDZ XJTI QKBI USMI AXEI JFZX XQWJ IBXM PCBW
BAFH OAXG BPBD KASZ LQOY ASNA ZPPC UTVF ISQZ HILW
GYCJ AUQV UGXH VQWW OXMX WEBW XSBQ EGZM OSKQ OYWK
MLKZ YPRC MMHW EHEQ DYJU XLVA CJPB SSVH WFPB WKGS
SAVE XVMF ZGCI SPEJ EFHD RXBS LZDX WYUW UWHB WTJG
ZFJM MTUV HHEZ GVKU IRON NNUU VWFD XZGW BUHB KSSG
VBHP KCQN CJCP GILT KZZK FMLW YVNS OXYJ AXQF RNML
NRYI RLIP CYPC WHQI YLUS JDDH HIDZ OWMH HHML CXXG
BJPY HWFI CMGP XGBP LBHZ ESGG XATC ERAY OSFN MKBP
WWKA PXHU UVIG ICOZ PDLW IMYC XZZH YQUV GUPM WYUE
RYPG TMWO GMTC PMXK ZLPS ACWW UFKO SJST OHGW AXOL
JNBX SQUP UIAP OMLU THXE LCGR HVDP HYLW HQXU OJMO
```

```

NCWD WIYK BZXM RPBO SYGI ZZYF AKCQ HVCN OHBT XEGI
GHXE YBZL OHSB DDZF AZPB AQVL AMCI TBFN TBCS ZNQG
FRCD WZRB KJEL MWDC RDAB NQIH DJKQ PYYB GNLN GWWL
CNCZ NEDW FXXR XPTH LVWN ZCXB TNYH PQUG TNFV AXFE
KSGO HNZZ PKHK HYQL SWQL RWXV LVAC WYWN IFBU JEPX
YSNJ IYXB JVNN QMEI FHMY KJTR YBAM HQBF FMWJ ZZVF
WUMS UPVJ BQUQ OJNP QWMM NABE YCCF RJTQ UHGL MMYM
KMIH VDKC EAOJ YCPB CDQV OEIL FLLO MGHG PPPA QOUS
UYIN MKBE HBGW VGVX ALOJ DGRD FBUQ XYJQ AYHV EKZY
SUJG LADI LTQT DXFZ AHLY LZWZ TBRJ BGKZ NRAB ZKXH
YYTB POXJ MXCP HDPR FCUG CSRB TQQY ZTBH ZZKP XEBO
TLZN XLOY ZSSV MQIU NVMP PSOG HBJZ CTPD EGXV AGWQ
MJBZ AMSI DYKS TZGB LPIW RMKO ZRYY OLTO YKKB YZWJ
EDDR RRLY WZRA PKNF IJGF SBJD IOQW BUCF PUEQ MYOZ
DBVQ GYUB CQJX VCSA TQCY XSUL RFNL ODFC KTKD CCIS
AZFE MHPR SAPV CGFQ HCAB JVBK CSWW CERU UOJX FERO
FGYS PIRG MDTP YWWV BOIK LOBZ QQON QCSN YZPM EEZP
NAJC TDLK VMAL VMRT OIHJ YKGL AYGM YCJE RKZD BBKE
QFNC IIEG JXMQ BMAS KJVG GITP HUTX YTMR YGIA YTHP
DRAF OZEG IHJV SARO HTPJ IVRN XGTJ EXJI MWBK CYCH
NOPN SVMM FRZH RAET IYKA GAKC HFVX BUTT JGGK HKWW
VKBY POVG IYIC KOZA ZDPK VXXE TZZR AJMQ RAVC FJIC
QNNR KJLG ZFMW QOQP KGFT DULW DWWP LKHB UIUP QDCE
CGOG IAIC UCWW GESB YSDW FAFX DYAF QGVZ JBVV VRAF
NAYL JQTN XDYU KANE WKZJ FVAI MSGZ PMKV SRAS PRPD
UVXV NXJI VNAY PVQZ WBPQ FXRG AOGP JXII CPFM ZKWP
IHQY KHOK IHGQ KTQH QTMD MWLG YUUG BOVY QBHG VKXF
LDUG LSWX JAQY SAVN PYFR CCLW OPPB OOMV IHQT SCKN
JTSL ZQLC ISKC LPUI NKJF CQZZ NEDW WXYI SBCO KVPY
RHLV CHMA BOCX GZXT TCHA FJGC GFGU LEGB GESE LXUQ
LZZO BIIP NLYI HDEC OSUH JJZX EPZP PCWP XAVT VNWG
AJQU HXCC ATXZ FMKD NFXC CCSX TNON RNRQ KJQB SODI
HOZQ HCUQ WQJQ YCYD BXHK NSQK RUOM NOCE IXRL PQHZ
NKAI TAGS ZFSI VQHG ULJV PZPJ KEBX ZXXT UCWX MXBR
MEPQ APOG COGY PJVO MSVW CWOK JYKV MNRB DZTO GATI
XXPN ZHLD OFLX LHWY SFJF IFCZ PDYG BHFQ YCRB GCHA
JAGX PNTK RFOL RWRD ZHPD NOGG JJTO PXPQ RWPU JTQY
YUIE GQJD YCYC MLHM FZLS HNFV QCUY FBCE XXGO FJPT
CQPS YHOD AVPR ALGP JPKB FFDH KLGS ZGXS KYUX BWSX
YPSY MYFY NWYO BNJY FDLR UPNV QRMU YWET RWEF PMCS
SKTK KNVO FPWB QBXA ENXM KMOZ SCPJ SDYX JFCV XPRJ
LDFQ DNCV VMXH WFLH CYIJ XHFC CYAD QRQO AGKT OXGL
VGJC ZFQE MTNZ OBZT AMMC RZMP SFEP YNZO LOIA QSPE
QRVQ LTEM DPZS RJBP YJEV WROV HRFO WJPS BENC AUJH
EDWS GFGJ RNYL BBKG KJXW VLXI EKQL LTUB PSNG XJRR
JZXP PLYP QPVS AXEM QSDI MFOF RMKC PMIU BGCI KMHU
VZRL QRAJ AEUP JQMY SUTY SYCQ FWYQ VDXA PXJV KQBS
HYDU QEUR SZUA FWQF OQDT DUFQ VOKW RWAC BFIF XZNM
VPWU ADXS LNCV LBAK OJNZ JMFO XQWL TZDT SGRB OTKG
DVMA XSQB BDNJ PPGB GIUN JGSV YFER AVPS WDYL XVVA
ZLQB GIBL TGJH DZNH OFKM DXMP VYQG GHIC IZXJ FDBA
XJUL WOZZ XUGC TTHW YGKI NSXI ZREE YYWR IKCS ONQE
NJTZ FZWC ATSM MQLA VLNN BCCS ZBKK FEHZ GTYE ZYVC
XNYB RWXG RHQT RDWK FHJP THQA YPEG ZVQW ZXNX WMRO
YZMU HDDC BAWN MATS XAWM GHDJ TSEJ BOAE EVWU YYMD
YJQU NFEC QLWD WTKG CBTN VPLA DQDY HAYB UPLJ ZHTB
XMLB MTUP AMWM IONR MJRB XOYL PYZU JTRY KATU UAOZ
MLBV JZLL BDYV MXYR OQGX GRUK STVT MLSX OLZJ IEJT
NLRV HNUV PIKX VOBQ EXMU BHZK FHNC UNCN WVEY BHRU
XXSW IXFC DAUC ZYLR IECI ZVDW CTUN FVDE RJZP TRQY
EHMF MDDS DJTE BVIO ZFVO YPXM KKXA SYNS BFFW HOLQ
FZVA IFHS PZJB YAYH CMMD OTLD GEVL LLXJ SEKD WFAD
HYYG ZIIV ZOOM WANE XHND CDTU CGAV JOKK MTLE YDDW
QEFP NIVT CCUL DENX SCSU BPUH GIIT TTCN NCQP UBKC
YCXX PWVO TGVS KNBP GDMS DHIG OMQY IVQH IVCG CARZ
HRXX WSRU WWKO EOQC QKQF GRCF KOQD NRUH MNCT BZGP

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JPGH BYXW RMVV QOWW PNCX BFQM BNGV HLXX WFZE WSAY
SMXY VWOZ LNPV HVHL MSAF EZIJ EVHY EWSI SHUJ VKIG
SNZI IHGV BIGX DELG NXTU KEWJ LJRG YLES JURS TGMN
JUBX QSYU BKAQ YMGM BPBI PCMK UYCY YPIS IVDH NNLO
KNKT MJIE QTKQ IXQF INFF ZOFV NBZS UQLP ZNLF DXCI
FPRW MIHU RRNL VNFQ DCEB MNZH AEAU FRND JIIP KIEJ
WPPI ZPON TRZN BNNT DANG VYEJ OGDY SZVG ZEPZ ZOME
JJJT QHZZ POGO NMYJ DIIJ PQEM YTHF QCFW WTDQ EQML
CENZ ZOOR FUXJ FLZT YEAQ YOEK QTBQ JBDC DLFV PTDA
FFFU DYUS MXPV HBGJ EFJZ POMU HMSJ CYVY QOBV XDEW
LWKV GZXU LHPR WTPV BKMV VESX PFNS PUEN VZCU OEKV
CHJR YODV METT PSYW WFMS EEZV YGJK DBNA KHER SSXE
BZVY NLPL FVRV KUBC YOTP TMOT BCXV GGJU YGYH GVBZ
CLAR HADF VAOX HPGH PVEB XVVY BGTL AMPC UCIL UMCX
FPCP GHGM MXYV IXVH HVTI MUPU XDCN GISL LLMI JUEH
RTMV CKFE ULKD NLYP ZZQC VFUI IWZG TGMH LPSI TEJD
UZTP GBVI XEDU QNCL RIAR UNQA LHAF RFLJ JSTJ BXHL
QSQV XQZA KLLK YGKX BTHY CAPR PORQ DAKA NMYW KHJE
EIGX VGKG NXGL FTEA HEFX BNQU GIKS TJJH CNAJ XQQK
XRUV ZKAA YMTX XUHN RPBQ RGOI QTUJ XEIU NLBZ OFEV
PIHB UUMW NHHH RACR LRIR UQWQ NVYS QAQI LWQL RTLR
XWHD WNSF AURV FPMI DAKA LPJL SZDH RJRV QNEY OTGQ
AZOF SYPB QMSK GSUK HINN TNIJ HRWI KYPM VRBV CLQL
VINE VGYN CEAQ UWIE QJUI SYGU WEJJ OGTA MOAQ XDMU
XQML BINO TTYV JYEE LXOW NXXV ZSLO GMLO ZQHJ OTLZ
KVXC GNDJ MAQE IXWS SBPI NZST VOUJ EFJC LWVP TGVR
IBOV DRAP VLDQ AKQB DTJM YNGM GBIY IQZD BBRN KHHB
ZOEY LFCM FULG FTSU DFIW BSDN RTDF PJWM FKJH FCLR
PYCN FPYR AJIB SEGV DSGY TZPD CYMH BCLV FXFG DFSU
HUXM SNCN PGCS QOOG BEET WFMV BLUK ZSVA PWVX RHDE
CPUS CFOV ZGKV QSNX ZORB SIGH PXQD WYPY IPQY QTHE
FPLK LHUK BGAG CIVW RQCB MLCB TJKQ XXGL GOTL PXQN
UKWH UASG CXBO NEWQ PMPB DUKA RQCB LJZW PXUK YLWC
GHCH EZPR GFFL DGQH TRUQ FFGG YOQX ZQWY UVMX ZIZH
PVTI HKPK RXVO ZHBP ZFIM NJZZ BLRY YBJE DLNX RVBG
ONOK YHLR QENX TSXX PMQP YYAD PYIH PRHA ZRSX WGTW
ADKA GLBT VKPH WNVS XJNE SNNT SRCK FAVZ HROC MSHS
WVRS GJXY ZELC OYWT JXOU VVPM AWSV ZXJU WNSE KQCJ
EUSP UWTG DLYY QWTM UWAC OFIY BFCY JKXS TSYW EGSG
DKME FPDQ DLGR UPFN VWUD AYPC UFCP AVIV QMWU XONL
CGSC EFWR DRUG YQHM SLCT IQOS WRBF WSAM RXWA AIVO
GCPY SCLZ DWTX RLDZ MLST HHJW XRYD NYMK SIIE DXCD
YHLC JNLN HNVH YBQW FFFF KSOM VPOR OQUA ODVC NZEM
QYPS VTAK ZNXX PWOL QYKB JGDP WGHK IRAS HCID LLGC
VFUW TMUF LOMK NOWC ISNB OUQR ZRVW KUUU BYEO EYDF
JVDV NGXR BYEE RNJA XWVG IMXR BQFR ZZAK FPNQ GNOI
OALF VWOK PTRF MFBM PHRL MJAT CHJG CEYD NJVE WJBR
CGPS LONH XDHN EANP SMYL VJNJ FWVI NWRH LGBB YBHW
PDGI FWEX BCCC ZPOQ RSAY OUGR QWPX JOHS NAHO EFVG
KIKV XCAD DTQC VUUQ RCJC LNSU XKUM PRKM HPQF GAQI
CFGD TOKS LAMZ KAJG HSER HUNH PMXM SGFE VBAU SOYB
GXNI APKW FNEW HGHP DXEP GPIU CNIB DDIG YCCE TOWB
OSJZ CHQU ALIO XGUI APYG UDZD QUZZ FBAS EAKX CXSU
EQDT EUIP SWUT BESM BDFG FQZB SITO ZHEB AIPO QISJ
BBZS BWHQ CTHR PBKK DRPQ UTTG RXJJ SKVC OJUW MLRR
IEUK GFTL NUUZ BMIF GYWI BFUK MMXJ ZSTL VFLR FBFR
OTZA SJBH FABD RXPV UCTN KFMM CPWH ZVYB PDCR TZOM
XTQU PKJT MGVE UFKC XZJY KOVN TUDL YUML USPF GZVZ
KCJX YQRY YEHF SEKL JPRS SOTJ NPUK MHGX CNCD VXDW
NMTV BYMA RGWM UGBH POWX XYEW YPXV FKHS ETBV YBUJ
PXGE GCOL UOHY BIPX NIKP VDBR OPGQ XSSX PEWQ PCXZ
NITG WNRV QVSW SBWY FWNC CYZV SNFL SQUP UQIA QDXM
DYFV YDPK YXWP YQFQ SKQA RWGZ JJJJ YJBF VTSU HOVW
EIRO RIKC MPPF MMQY VDOU QAEJ DHUO PINZ BPQD ENDH
CDJG THOB DERH RMZJ BKDK PDFF RPQN UQRG KNFC NMPE
NCKM NPZJ RKGM RKPI WNXS PEGG BRPL HQUC KVJJ UMFM

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KVZR YOGC YJXM UZVN DRUK WBHQ XZPH DPYI LQUV KBPP
TYNF UVZZ YQWO JTSN IFNQ LPLR JZEJ ZZQJ LNKQ OKQT
UMLF QVOZ WSLP THJF DJUC KQEX ISCX QCEP VCQC NMRR
QLXZ XFOP FZVP ZXGJ RQSS BLPS JDZZ PLSW RCPC MCNG
FCJD PQIO GNCA MYHA GOCF BVZE REZY WIDC DRYX GQJW
VLKC KMXZ KTRT SQDH EDKW GDBT KBRX CZRK HDYU HVIF
WNFX QGFT OYQU NQLB SBRQ QFGR FTXF HCCV XQWW CZQN
SZCG VFMZ EZGD XSWI WFZU SPBQ CCOJ GLZI GRUY DREP
LRVR OFSN KKVJ BANW OINR ONFF OUKA BKPM EPWJ FJZX
AJKZ CNCR TAMK FXHY ZCAH GHTU ZTKF PCBR HPEK VXCD
HXKB QTVH CHWU ZJCX HRPT UMVC UBIO XSGO AINB APYK
CZJS DRBV JCLT QFNR YQRS FKXT TBWY LDVC DJNJ CVED
TXQB AVIW VVDB WZHY WGQQ ZZYF NVXW OPBQ CRDD BDLW
SCQG CGXD UTMB RBGA VMWJ MFEE QHBM PYDL YMGC IJKF
OUES HWWO ELRO XHHF CLSV QBTT BPTO NGKM DDWB FGAP
DEYU JNHR FQAY DPTA TRND NBCI SBHR OASR CMJU HPXQ
KLDD IVHA SHNT HGPZ PXWU MGYL JWLF DSWE SMQN AQOM
MIQX MIPV PWPA YBMN VWSQ WGJX EPHY ICAB MEMB NMVI
XLUO XOUB SOOY PXIS CBQP ZDNU PDYU JQUA GXHX KSJK
FGUJ YQDW EIXP VBSE IANY MWIR ITTP MUVF SUYV XUAR
YKSO JEVZ AMFS QZPY TKIN JZVO IGFK LNGL MNUK RIEU
CKVQ BQVL VIWT MZKP VKZZ FRBX CJXC ORRD SHMO LWBQ
OSVO VOVN XUIZ CNLJ FWVF QSNB NVXN CHFG XYYO GVLU
EDZK OWBC AWOX XTSI XRQO XSHS NBJP CQKW XPLB BBJT
QLYS NBLB QXSN UXPL KSYF BFUH INXV ZPTN YQTF RQCO
AYDG ZFVB MCAG ETBM XZYJ EVEP BCXU ZVQJ PEBD DZUX
JFNL VMZY CKXS QRIN MZWL KUVY VPEJ UHLR REWR BUGH
LCGI TBPV VEAG ETCM VBDR CFVM AZTS JLPY OJPM JTKB
NKKP MJXW FFFA RAYR YSYX HCBH QWUY LKBG ZIKI ICYH
LXOD KHYL HEWJ BNXQ RGXN GWUD LKPJ XVXD MHPE ORRL
JTZV MONB YISX SVEE CHXG JNLY QBYA WPQZ STDO BEJF
JEMV WPFO BJZP LIDL GGOG CPVR JQEP OLNI HVCR QNWQ
LQHJ BXFD F

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Analisis Modifikasi Mesin

Langkah Penyelesaian

Artefak

Plaintext Hasil Dekripsi

Kunci / Konfigurasi

Hasil *Decrypt* Sesuai Format Sumber Asli

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